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Default and Recovery Rates of Corporate Bond Issuers, 1920-2004

Summary

This report is Moody's 18th annual survey of global corporate defaults and recovery rates. Moody's reviews the default, recovery, and credit loss experience of corporate bond issuers for 2004 as well as the historical period since 1920. Briefly, we find:

- A total of 34 corporate bond issuers rated as of January 1, 2004 defaulted on a total of USD 16 billion in 2004. Including issuers whose ratings were withdrawn prior to January 1, a total of 39 Moody's-rated corporate bond issuers defaulted on a total of USD 16.6 billion.
- As a percentage of issuers, the default rate for all rated corporate issuers fell to 0.7% in 2004 from 1.7% in 2003. The default rate for all rated corporate issuers fell to 0.4% in 2004 from 1.0% in 2003 as a percentage of dollar volume. The speculative-grade issuer default rate fell to 2.2% in 2004 from 5.2% in 2003 on an issuer basis and fell to 2.5% from 5.7% on a dollar volume basis. No investment-grade issuers defaulted in 2004.
- Defaults were concentrated in the lowest rating tiers of speculative grade. Issuers rated Caa and below defaulted at a 12.3% rate in 2004, while Ba-rated issuers defaulted at a rate of just 0.2% and B-rated issuers at 0.7%, considerably below their long run averages.
- The telecommunications sector experienced the highest total default volume for the fifth straight year, with 3 firms defaulting on a total of USD 6.1 billion; the metals and mining sector experienced the highest number of defaulting issuers (4).
- Moody's forecasting model for its global speculative-grade default rate predicts a nadir of 1.8% in the second quarter of 2005. By the end of 2005, the global speculative-grade default rate is expected to rise to 2.7%, suggesting an average 3.6 corporate issuer defaults per month.
- Aggregate credit quality improved in 2004, as rating upgrades exceeded rating downgrades for the first time since 1997. The downgrade-upgrade ratio fell to 0.7:1 in 2004 from 2.4:1 in 2003; 14% of outstanding issuers were upgraded, while 9% were downgraded.
- Rating performance improved in 2004. The median rating of 2004 defaulters was B2 five year prior to default, lower than the historical average of Ba3. The one-year accuracy (AR) ratio increased to 90.7 in 2004 from 88.8% in 2003.
- Recovery rates of defaulted corporate bonds rose sharply in 2004, with the average issuer-weighted senior unsecured bond recovery rate reaching 50% for year. Defaulted bond recovery rates increased across all seniority classes.



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Introduction

This report is Moody's 18th annual review of corporate defaults, recovery rates, and ratings performance for 2004 as well as the historical period since 1920. Each of these topics is addressed in its own section below. In addition, this report contains two special appendices. Appendix A contains detailed default, recovery, and rating transition data spanning the 1920-2004 time period. Appendix B contains important information about the methodologies and data used in this study.

Corporate Bond Default Rates

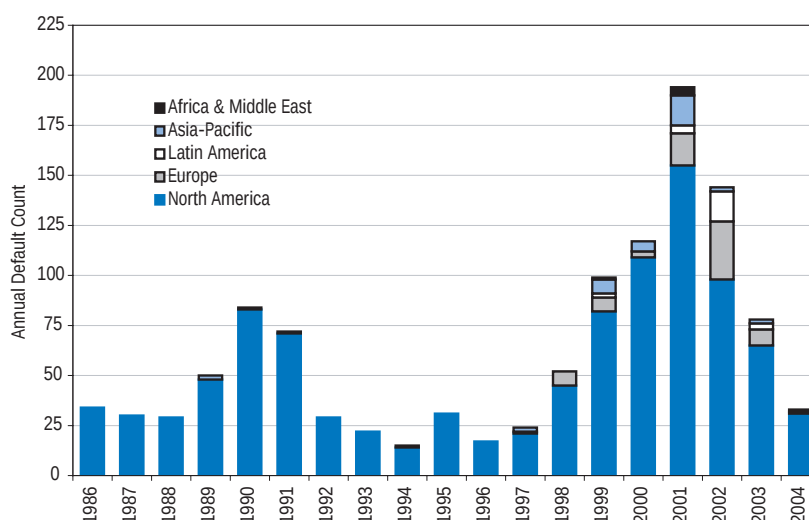
Corporate credit quality experienced considerable improvement in 2004. The number and dollar volume of Moody's-rated corporate bond defaults declined for a third straight year, and fell to half of 2003's level. A total of 34 corporate bond issuers rated as of January 1 defaulted on a total of USD 16 billion in 2004. Including issuers whose ratings were withdrawn prior to the beginning of the year, a total of 39 Moody's-rated issuers defaulted on a total of USD 16.6 billion (see Exhibits 13 and 21 in Appendix A). In comparison, a total of 80 corporate issuers defaulted on USD 34.2 billion in 2003.

Sixteen defaults occurred due to missed interest payment; 12 defaults occurred due to bankruptcy; and 11 issuers defaulted as a result of a distressed restructuring in 2004.

Thirty of the 34 defaulted issuers (88%) rated as of January 1 were U.S.-based. Two of the four non-U.S. issuers were based in the U.K. (Avon Energy Partners Holdings and Equitable Life Finance PLC), with one Canadian (Hollinger Inc.) and one based in Israel (BARAK I.T.C.). Moody's recorded no rated bond defaults in continental Europe in 2004. The additional five Moody's-rated defaulted issuers whose ratings were withdrawn prior to January 1 were all U.S.-based. Yukos, the Russia-based oil giant, defaulted on nearly one billion dollars of bank loans. Although Yukos carried a Moody's issuer rating at the time of default, it did not have any rated bonds, and is not, therefore, included in Moody's bond default statistics. Exhibit 1 shows the annual count of defaulting corporate bond issuers by country of domicile.

Level 3 Communications, Inc. was the largest default in 2004 with bonds totaling USD 4.7 billion, and was one of only four issuers with defaulted debt totaling over one billion dollars. The other three defaults in excess of USD 1 billion include Trump Atlantic City Associates, RCN Corporation, and Avon Energy Partners Holdings. The average size of default in 2004 was USD 471 million.

Exhibit 1— Annual Corporate Bond Issuer Default Counts by Geographical Region, 1986-2004



For the fifth straight year, the telecommunications industry experienced the highest total default volume. Three telecommunications firms defaulted on a total of USD 6.1 billion. The metals and mining sector experienced the highest number of defaulting issuers, with four issuers totaling USD 584 million. Exhibit 2 shows the industry sector distribution of defaults in 2004, as well as 2004 default rates for each industry category.

Exhibit 2 - Distribution of Rated 2004 Corporate Bond Issuer Defaults by Industry Sector

Industry	Issuers	Volume*	Default Rate (% Issuers)
Aerospace & Defense	1	\$140	2.00
Beverage, Food, & Tobacco	3	\$270	1.49
Chemicals, Plastics, & Rubber	1	\$275	0.63
Construction, Building, & Real Estate	2	\$420	1.57
Transportation	2	\$1,271	3.64
Diversified/Conglomerate	1	\$489	1.69
Electric Utilities	1	\$1,138	0.35
Electronics	1	\$331	0.54
Forest Products & Paper	1	\$330	1.32
Healthcare, Education, & Childcare	1	\$175	0.89
Hotels, Casinos, & Gaming	3	\$1,937	3.30
Insurance	1	\$672	0.33
Metals & Mining	4	\$584	3.92
Miscellaneous Manufacturing	1	\$100	2.08
Consumer Products	1	\$118	0.87
Oil & Gas	1	\$250	0.27
Printing, Publishing, & Broadcasting	2	\$916	1.14
Retail	2	\$247	0.02
Telecommunications	3	\$6,098	1.26
Textiles, Leather, & Apparel	2	\$257	6.06
All Industries	34	\$16,019	0.72

* USD millions

As a percentage of issuers, the default rate for all rated corporate issuers fell to 0.7% in 2004 from 1.7% in 2003. As a percentage of outstanding debt, the default rate for all rated corporate issuers fell to 0.4% in 2004 from 1.0% in 2003. The speculative-grade issuer default rate fell to 2.2% in 2004 from 5.2% in 2003 on an issuer basis and fell to 2.5% from 5.7% on a dollar volume basis. No issuers defaulted within a year of holding an investment-grade credit rating in 2004.

Defaults were concentrated in the lowest rating tiers of speculative grade. Issuers rated Caa and below defaulted at a 12.3% rate in 2004, while Ba-rated issuers defaulted at a rate of just 0.2% and B-rated issuers at 0.7%, considerably below their long run averages. Exhibit 3 shows default counts, dollar volume totals and default rates by whole letter rating category; Exhibit 4 presents the data at the alphanumeric rating level.

Exhibit 3 - 2004 Corporate Bond Defaults by Whole Letter Rating

Rating as of 1/1/04	Global				United States			
	Default Count	Volume (Billions)	Default Rate (% Issuers)	Default Rate (% Volume)	Default Count	Volume (Billions)	Default Rate (% Issuers)	Default Rate (% Volume)
Ba	1	\$0.13	0.19	0.00	1	\$0.13	0.30	0.00
B	5	\$0.74	0.65	0.20	4	\$0.56	0.68	0.24
Caa-C	28	\$15.15	12.33	12.46	25	\$13.22	12.89	11.86
IG	0	\$0.00	0.00	0.00	0	\$0.00	0.00	0.00
SG	34	\$16.02	2.23	2.48	30	\$13.91	2.69	2.50
All Rated	34	\$16.02	0.72	0.43	30	\$13.91	1.14	0.60

Exhibit 4 - 2004 Corporate Bond Defaults by Alphanumeric Rating

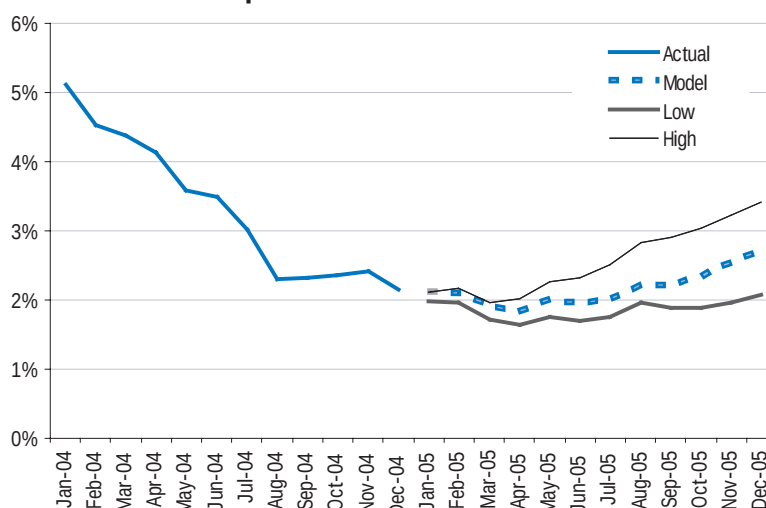
Rating as of 1/1/04	Global			United States		
	Default Count	Volume (Billions)	Default Rate (% Issuers)	Default Count	Volume (Billions)	Default Rate (% Issuers)
Ba1	0	\$0.00	0.00	0	\$0.00	0.00
Ba2	0	\$0.00	0.00	0	\$0.00	0.00
Ba3	1	\$0.13	0.47	1	\$0.13	0.77
B1	0	\$0.00	0.00	0	\$0.00	0.00
B2	2	\$0.36	0.66	1	\$0.18	0.42
B3	3	\$0.38	1.52	3	\$0.38	1.89
Caa1	10	\$3.20	8.66	7	\$1.27	7.37
Caa2	4	\$6.59	7.08	4	\$6.59	7.77
Caa3	5	\$1.32	20.41	5	\$1.32	23.81
Ca	8	\$3.92	30.77	8	\$3.92	36.36
C	1	\$0.12	22.22	1	\$0.12	22.22
IG	0	\$0.00	0.00	0	\$0.00	0.00
SG	34	\$16.02	2.23	30	\$13.91	2.69
All Rated	34	\$16.02	0.72	30	\$13.91	1.14

Credit quality measured by credit rating changes¹ also showed significant improvement in 2004. The downgrade-upgrade ratio fell to 0.7:1 in 2004 from 2.4:1 in 2003; 14% of outstanding issuers were upgraded, while 9% were downgraded. Notably, rating upgrades exceeded rating downgrades for the first time since 1997 in 2004. Exhibit 29 in Appendix A details the annual credit rating change statistics. Exhibits 30-37 in Appendix A present complete credit rating transition matrices.

2005 Default Rate Forecast

Moody's forecasting model² for its issuer-weighted speculative-grade default rate predicts a near-term decline followed by a steady increase in 2005. The speculative-grade default rate is expected to reach a cyclical low near 1.8% in the second quarter 2005, rising to 2.7% by the end of the year. Exhibit 5 shows the model-generated forecast, as well as a high-default and low-default scenario. The default rate predicted by Moody's model suggests an average monthly default count of 3.6 issuers for 2005. The high default rate scenario assumes that 4.5 issuers will default per month, while the low default rate scenario assumes 2.8 issuer defaults per month. Based on these scenarios, the default rate can reasonably be expected to end 2005 in the range of 2% to 3.4%, well below the average historical default rate of 5%.

Exhibit 5 – 2005 Speculative-Grade Default Rate Forecast



1. Rating changes are measured using estimated senior ratings, not individual bond ratings. Please see Appendix B.

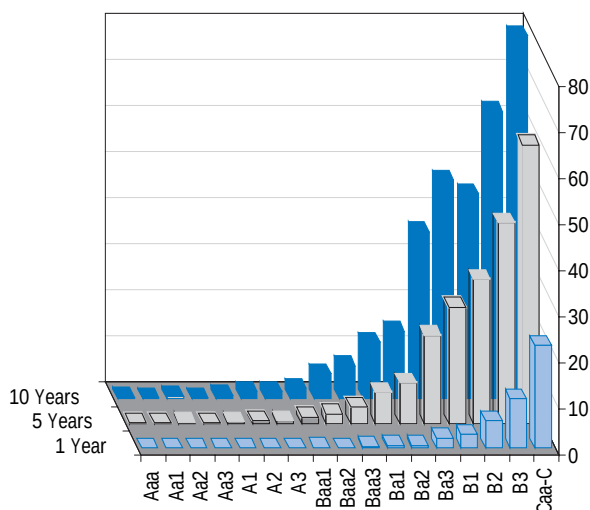
2. Keenan, Sobehart and Hamilton (1998) contains a detailed description of Moody's forecasting model.

Expectations of a rising default rate in 2005 (and beyond) are driven primarily by credit quality factors that are already taking shape. The number of newly rated, speculative-grade corporate bond issuers grew by 37% from 2003 to 2004, representing 46% of all newly rated issuers in 2004. As speculative-grade issuance has grown, credit quality has declined. In 2004, issuers rated below Ba constituted 39% of new issuers, the highest level since 1998. More worrisome for future credit quality, however, is the fact that nearly 16% of all new issuers in 2004 were rated in the Caa-C range, a historic high proportion (see Exhibit 24 in Appendix A). Moreover, when measured as a percentage of dollar volume, the Caa-C rating segment represents over a third of outstanding speculative-grade issues. While Moody's currently does not expect the default rate to rise sharply over the next year, the high proportion of Caa-and-below rated issuers will eventually put upward pressure on the aggregate default rate.

Credit Rating Performance

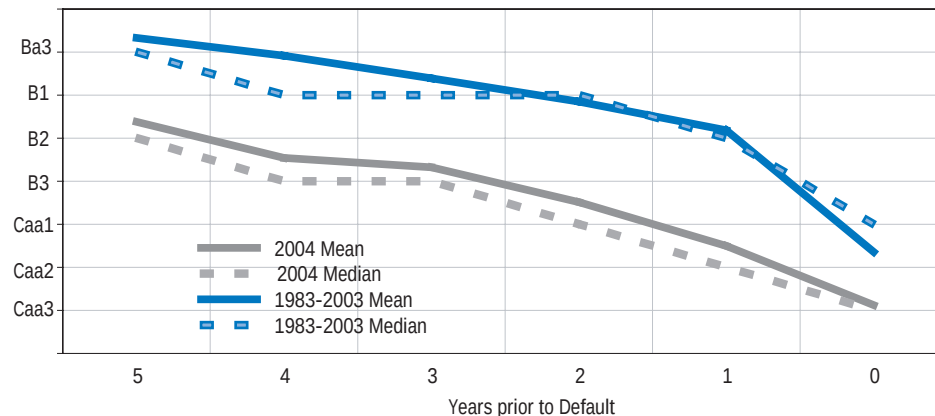
Moody's credit ratings have been highly correlated with subsequent default rates over investment horizons as long as 20 years. Exhibit 6 shows that average cumulative 1-, 5-, and 10-year default rates between 1983 and 2004 increase with lower credit ratings over each time horizon. The tables of annual and cumulative default rates in Appendix A show more detail for various historical time periods and investment horizons.

Exhibit 6 – One, Five, and Ten Year Default Rates by Rating Category



As the section above showed, default rates were quite low across all rating categories in 2004, and concentrated in the lowest tiers of speculative-grade (Caa and below). In the expansionary phase of the credit cycle low default rates across all rating categories are to be expected, as was the case in 2004. Exhibit 7 shows, however, that issuers that defaulted in 2004 possessed lower ratings than the historical average as long as five years ahead of default. The median rating for 2004 defaulters was B2 five years prior to default, below the historical average of Ba3. In the year prior to default, the median rating was Caa2 in 2004 versus B2 for the historical period from 1983-2003.

Exhibit 7 – Median and Average Ratings from Five Years Prior to Default



Also of interest is the power of a rating system to properly rank order non-defaulters as well as defaulters. Exhibit 8 presents a set of ratings performance metrics that address this concern for 2004, 2003, and the historical average. The annual accuracy ratio³ increased from 88.8% in 2003 to 90.7% in 2004, and the five year accuracy ratio edged higher to 70.6%.⁴ The large (greater than two notches) rating action rate increased in 2004 due to a number of multi-notch upgrades in Asia. Overall, the figures show that Moody's ratings showed improvement in their ability to differentiate defaults from non-defaults in 2004.

Exhibit 8 - Rating Performance Metrics			
Metric	2004	2003	1983-2003 Average
1-Year Accuracy Ratio	90.7%	88.8%	84.4%
5-Year Accuracy Ratio	70.6%	70.0%	71.8%
Average Rating 1 Yr. Prior to Default	Caa2/Caa1	B2/B3	B2
Rating Action Rate (annualized)	21.7%	21.5%	22.3%
Large Rating Action Rate (annualized)	5.6%	6.5%	4.5%

Default Severity & Credit Loss Rates

Commensurate with declining default rates, recovery rates continued their upward trajectory in 2004 after reaching a cyclical low in 2001. The issuer-weighted mean senior unsecured recovery rate for defaulting issuers rose to 50.1% in 2004 from 41.4% in 2003, and is above its 45% historical average. Average recovery increased for all debt priorities in the capital structure, as shown in Exhibit 9. Exhibit 27 in Appendix A presents average annual defaulted bond recovery rates and descriptive statistics by priority in the capital structure.

Exhibit 9 - 2004 Issuer-Weighted Recovery Rate Descriptive Statistics							
Priority	1982-2003						
	Mean	Mean	Median	Min	Max	StDev	N
Sr. Secured	\$57.4	\$80.8	\$94.3	\$40.0	\$100.0	\$25.5	8
Sr. Unsecured	\$44.9	\$50.1	\$47.0	\$15.0	\$95.8	\$22.3	33
Sr. Subordinated	\$39.1	\$44.4	\$48.0	\$8.0	\$90.0	\$25.7	7
Subordinated	\$32.0	NA	NA	NA	NA	NA	0
Jr. Subordinated	\$28.9	NA	NA	NA	NA	NA	0
All bonds	\$42.2	\$54.3	\$49.0	\$8.0	\$100.0	\$26.1	48

The rise in recovery prices of defaulted bonds in 2004 is due to several factors, including higher asset values, low speculative-grade spreads, and a decline in the supply of new distressed debt. Moody's studies, as well as academic research, have noted that when default rates are high, as they were in 2001 and 2002, recovery rates tend to be low, and vice versa.⁵ This negative correlation between average senior unsecured recovery rates and annual default rates is evident in Exhibit 10. A simple linear regression of senior unsecured average recovery rates on annual default rates for all rated corporate issuers suggests a high correlation, with coefficient of determination of 0.65. A similarly strong correlation exists between average senior unsecured recovery rates and speculative-grade default rates. If Moody's forecast of higher default rates in 2005 and beyond is borne out, Exhibit 10 suggests that recovery rates should contemporaneously fall.

3. The accuracy ratio (AR) is described in Cantor and Mann (2003).

4. Mann (2005) presents a thorough discussion of Moody's 2004 rating performance.

5. Varma (2004) is an in-depth analysis of the drivers of historical recovery rates

Exhibit 10 – Correlation between Recovery Rates and Annual Default Rates, 1983-2004

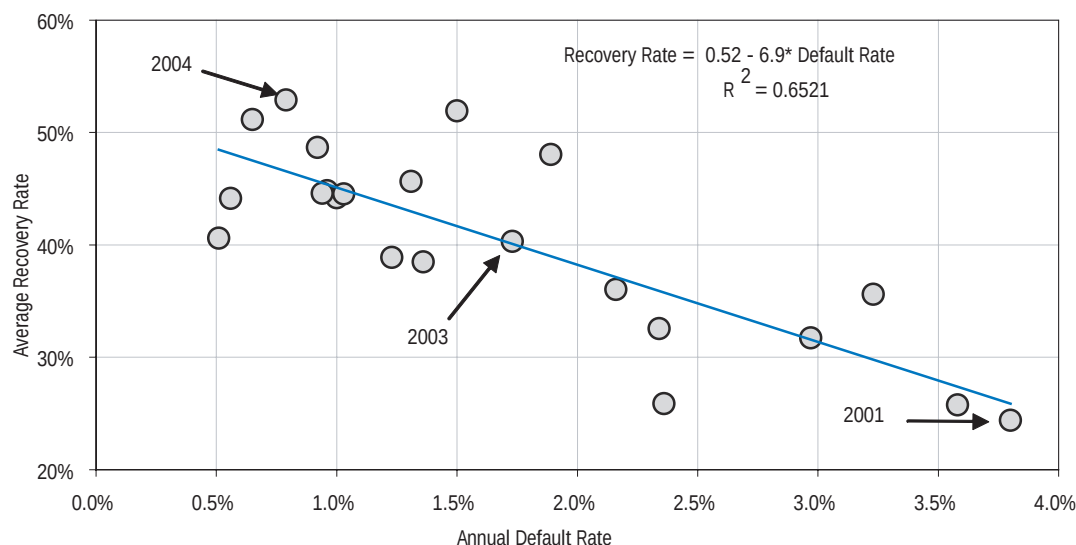


Exhibit 28 in Appendix A shows the existence of a long-run relationship between average senior unsecured recovery rates and Moody's estimated senior ratings. At least one-year prior to default, Moody's ratings properly rank order issuers based on recovery rates: issuers rated higher one-year prior to default experienced higher recovery rates compared to lower-rated issuers. Additionally, issuers rated investment grade at any time five years prior to default have experienced higher average recovery rates. While the difference between the average recovery rates for investment-grade and speculative-grade issuers is small at long time horizons, it becomes larger as the issuers near default.

Moody's credit ratings are opinions of relative expected credit loss, which is comprised of both the probability of default as well as the severity of default. Exhibit 11 shows that Moody's credit ratings have consistently ranked credit losses for investment horizons as long as five years.

Exhibit 11 - Average Cumulative Credit Loss Rates by Whole Letter Rating, 1982-2004

(Percent Loss per \$100 Par)

Sr. Unsecured Rating	Time Horizon (Years)				
	1	2	3	4	5
Aaa	0.00	0.00	0.00	0.00	0.05
Aa	0.00	0.00	0.03	0.07	0.12
A	0.01	0.05	0.14	0.23	0.32
Baa	0.13	0.37	0.59	0.99	1.35
Ba	0.77	2.10	3.76	5.28	6.91
B	3.58	8.14	12.17	15.61	19.51
Caa-C	14.18	23.79	32.30	31.58	53.06
IG	0.04	0.13	0.24	0.39	0.55
SG	3.25	6.66	9.78	12.03	14.88
All Rated	1.09	2.22	3.22	3.97	4.82

References

Cantor, R. and C. Mann, Measuring the performance of corporate bond ratings. Moody's Global Credit Research, April 2003.

Keenan, S., Sobehart, J. and D. T. Hamilton. Predicting default rates: a forecasting model for Moody's issuer-based default rates. Moody's Global Credit Research, August 1999.

Mann, C. The performance of Moody's corporate bond ratings: December 2004 quarterly update. Moody's Global Credit Research, January 2005.

Varma, P. Determinants of recovery rates on defaulted bonds and loans for North American corporate issuers: 1983-2003. Moody's Global Credit Research, December 2004.

Appendix A - Statistical Tables

Corporate Bond Defaults

Exhibit 12 - Moody's-Rated 2004 Corporate Bond Defaults

Company	Country	Default Date	Total Face Amount*	In Jan. 1 Cohort
aaiPharma Inc.	United States	4/1/2004	\$175	Yes
American Restaurant Group, Inc.	United States	5/1/2004	\$162	Yes
Applied Extrusion Technologies, Inc.	United States	7/1/2004	\$275	Yes
ATA Holdings Corporation	United States	10/27/2004	\$398	Yes
Avon Energy Partners Holdings	United Kingdom	1/15/2004	\$1,138	Yes
BARAK I.T.C.	Israel	11/15/2004	\$183	Yes
Better Minerals & Aggregates Company	United States	6/9/2004	\$150	Yes
Continental Global Group, Inc.	United States	4/1/2004	\$120	Yes
Dan River Inc.	United States	3/31/2004	\$157	Yes
Equitable Life Finance PLC	United Kingdom	12/23/2004	\$672	Yes
Falcon Products, Inc.	United States	12/15/2004	\$100	Yes
FiberMark, Inc.	United States	3/31/2004	\$330	Yes
Foster Wheeler LLC	United States	9/21/2004	\$220	Yes
Foster Wheeler Ltd.	United States	9/21/2004	\$200	Yes
G+G Retail, Inc.	United States	3/15/2004	\$107	Yes
Haynes International, Inc.	United States	3/1/2004	\$140	Yes
Hollinger Inc.	Canada	3/1/2004	\$120	Yes
INTERMET Corporation	United States	9/30/2004	\$175	Yes
Interstate Bakeries Corp	United States	9/22/2004	\$130	Yes
Jordan Industries, Inc.	United States	2/18/2004	\$489	Yes
Level 3 Communications, Inc.	United States	12/2/2004	\$4,663	Yes
Metallurg, Inc.	United States	6/1/2004	\$140	Yes
Mrs. Fields' Original Cookies, Inc.	United States	3/15/2004	\$140	Yes
New World Pasta Company	United States	2/15/2004	\$160	No
Ormet Corporation	United States	1/30/2004	\$225	No
Pegasus Media & Communications, Inc.	United States	6/2/2004	\$81	No
Pegasus Satellite Communications, Inc.	United States	6/2/2004	\$796	Yes
Protection One Alarm Monitoring, Inc.	United States	7/15/2004	\$331	Yes
R.A.B. Enterprises, Inc.	United States	4/26/2004	\$80	Yes
R.A.B. Holdings, Inc.	United States	4/26/2004	\$73	No
RCN Corporation (Old)	United States	1/15/2004	\$1,252	Yes
Syratech Corporation	United States	10/15/2004	\$118	Yes
Tom's Foods Inc.	United States	11/1/2004	\$60	Yes
Trico Marine Services, Inc.	United States	5/15/2004	\$250	Yes
Tropical Sportswear Int'l Corporation	United States	12/15/2004	\$100	Yes
Trump Atlantic City Associates	United States	10/21/2004	\$1,300	Yes
Trump Casino Holdings, LLC	United States	10/21/2004	\$475	Yes
US Airways, Inc.	United States	9/12/2004	\$873	Yes
Wickes Inc.	United States	1/20/2004	\$43	No

* USD Millions

Exhibit 13 - Annual Corporate Bond Issuer Default Counts

Cohort Year	IG	SG	All Rated	Cohort Year	IG	SG	All Rated
1920	8	25	33	1963	0	3	3
1921	7	24	31	1964	0	0	0
1922	10	23	33	1965	0	0	0
1923	5	22	27	1966	0	1	1
1924	3	37	40	1967	0	0	0
1925	7	34	41	1968	0	1	1
1926	4	18	22	1969	0	0	0
1927	1	15	16	1970	1	26	27
1928	0	8	8	1971	0	3	3
1929	3	12	15	1972	0	5	5
1930	2	21	23	1973	2	3	5
1931	6	79	85	1974	0	3	3
1932	9	108	117	1975	0	4	4
1933	9	188	197	1976	0	2	2
1934	5	60	65	1977	1	3	4
1935	9	51	60	1978	0	4	4
1936	3	19	22	1979	0	1	1
1937	4	17	21	1980	0	4	4
1938	9	17	26	1981	0	2	2
1939	2	13	15	1982	2	11	13
1940	2	16	18	1983	0	13	13
1941	0	10	10	1984	1	12	13
1942	0	4	4	1985	0	16	16
1943	0	3	3	1986	4	30	34
1944	0	3	3	1987	0	30	30
1945	0	2	2	1988	0	29	29
1946	0	0	0	1989	2	48	50
1947	0	1	1	1990	0	83	83
1948	0	0	0	1991	1	71	72
1949	0	5	5	1992	0	29	29
1950	0	0	0	1993	0	22	22
1951	0	1	1	1994	0	15	15
1952	0	0	0	1995	0	31	31
1953	0	0	0	1996	0	17	17
1954	0	1	1	1997	0	24	24
1955	0	1	1	1998	1	50	51
1956	0	0	0	1999	1	97	98
1957	0	1	1	2000	4	112	116
1958	0	0	0	2001	4	180	184
1959	0	0	0	2002	16	129	145
1960	0	2	2	2003	0	79	79
1961	0	3	3	2004	0	34	34
1962	0	4	4				

Exhibit 14 - Annual Issuer-Weighted Default Rates, 1920-2004

Cohort Year	IG	SG	All Rated	Cohort Year	IG	SG	All Rated
1920	0.44	3.01	1.24	1963	0.00	1.13	0.35
1921	0.38	2.07	1.03	1964	0.00	0.00	0.00
1922	0.51	1.78	1.02	1965	0.00	0.00	0.00
1923	0.24	1.71	0.81	1966	0.00	0.40	0.12
1924	0.14	2.85	1.15	1967	0.00	0.00	0.00
1925	0.32	2.56	1.18	1968	0.00	0.37	0.11
1926	0.22	1.76	0.78	1969	0.00	0.00	0.00
1927	0.07	1.78	0.73	1970	0.14	8.78	2.64
1928	0.00	0.90	0.37	1971	0.00	1.10	0.29
1929	0.24	1.29	0.69	1972	0.00	1.88	0.46
1930	0.16	2.13	1.02	1973	0.23	1.24	0.45
1931	0.49	7.85	3.80	1974	0.00	1.31	0.28
1932	0.78	10.81	5.42	1975	0.00	1.73	0.36
1933	0.80	15.39	8.40	1976	0.00	0.87	0.18
1934	0.57	5.93	3.45	1977	0.11	1.34	0.35
1935	1.24	6.09	3.84	1978	0.00	1.78	0.35
1936	0.46	2.74	1.63	1979	0.00	0.42	0.09
1937	0.65	2.59	1.66	1980	0.00	1.61	0.34
1938	1.55	2.58	2.10	1981	0.00	0.70	0.16
1939	0.40	1.74	1.20	1982	0.21	3.54	1.04
1940	0.56	2.58	1.84	1983	0.00	3.82	0.97
1941	0.00	1.68	1.07	1984	0.10	3.32	0.93
1942	0.00	0.74	0.46	1985	0.00	3.67	1.01
1943	0.00	0.61	0.37	1986	0.32	5.64	1.91
1944	0.00	0.67	0.39	1987	0.00	4.23	1.51
1945	0.00	0.52	0.29	1988	0.00	3.59	1.37
1946	0.00	0.00	0.00	1989	0.15	5.79	2.27
1947	0.00	0.31	0.16	1990	0.00	10.08	3.64
1948	0.00	0.00	0.00	1991	0.07	10.40	3.28
1949	0.00	1.88	0.83	1992	0.00	4.85	1.33
1950	0.00	0.00	0.00	1993	0.00	3.52	0.95
1951	0.00	0.45	0.18	1994	0.00	1.95	0.57
1952	0.00	0.00	0.00	1995	0.00	3.33	1.04
1953	0.00	0.00	0.00	1996	0.00	1.67	0.52
1954	0.00	0.47	0.16	1997	0.00	2.06	0.66
1955	0.00	0.50	0.16	1998	0.04	3.45	1.24
1956	0.00	0.00	0.00	1999	0.04	5.65	2.16
1957	0.00	0.44	0.14	2000	0.13	6.14	2.41
1958	0.00	0.00	0.00	2001	0.13	10.58	3.82
1959	0.00	0.00	0.00	2002	0.49	8.45	3.04
1960	0.00	0.72	0.24	2003	0.00	5.27	1.70
1961	0.00	1.04	0.35	2004	0.00	2.23	0.72
1962	0.00	1.43	0.46				

Exhibit 15 - Annual Issuer-Weighted Default Rates by Whole Letter Rating, 1970-2004

Cohort Year	Cohort Rating									All Rated
	Aaa	Aa	A	Baa	Ba	B	Caa-C	IG	SG	
1970	0.00	0.00	0.00	0.27	4.12	20.78	53.33	0.14	8.78	2.64
1971	0.00	0.00	0.00	0.00	0.42	3.85	13.33	0.00	1.10	0.29
1972	0.00	0.00	0.00	0.00	0.00	7.14	40.00	0.00	1.88	0.46
1973	0.00	0.00	0.00	0.46	0.00	3.77	44.44	0.23	1.24	0.45
1974	0.00	0.00	0.00	0.00	0.00	10.00	0.00	0.00	1.31	0.28
1975	0.00	0.00	0.00	0.00	1.02	5.97	0.00	0.00	1.73	0.36
1976	0.00	0.00	0.00	0.00	1.01	0.00	0.00	0.00	0.87	0.18
1977	0.00	0.00	0.00	0.28	0.52	3.28	50.00	0.11	1.34	0.35
1978	0.00	0.00	0.00	0.00	1.08	5.41	0.00	0.00	1.78	0.35
1979	0.00	0.00	0.00	0.00	0.49	0.00	0.00	0.00	0.42	0.09
1980	0.00	0.00	0.00	0.00	0.00	4.94	33.33	0.00	1.61	0.34
1981	0.00	0.00	0.00	0.00	0.00	4.49	0.00	0.00	0.70	0.16
1982	0.00	0.00	0.26	0.31	2.72	2.41	25.00	0.21	3.54	1.04
1983	0.00	0.00	0.00	0.00	0.91	6.31	40.00	0.00	3.82	0.97
1984	0.00	0.00	0.00	0.36	0.83	6.72	0.00	0.10	3.32	0.93
1985	0.00	0.00	0.00	0.00	1.40	8.22	0.00	0.00	3.67	1.01
1986	0.00	0.00	0.00	1.33	2.03	11.73	23.53	0.32	5.64	1.91
1987	0.00	0.00	0.00	0.00	2.71	6.23	20.00	0.00	4.23	1.51
1988	0.00	0.00	0.00	0.00	1.24	6.36	28.57	0.00	3.59	1.37
1989	0.00	0.00	0.00	0.59	2.98	8.95	25.00	0.15	5.79	2.27
1990	0.00	0.00	0.00	0.00	3.35	16.18	58.82	0.00	10.08	3.64
1991	0.00	0.00	0.00	0.27	5.35	14.56	36.84	0.07	10.40	3.28
1992	0.00	0.00	0.00	0.00	0.30	9.03	26.67	0.00	4.85	1.33
1993	0.00	0.00	0.00	0.00	0.56	5.71	28.57	0.00	3.52	0.95
1994	0.00	0.00	0.00	0.00	0.24	3.82	5.13	0.00	1.95	0.57
1995	0.00	0.00	0.00	0.00	0.69	4.81	11.57	0.00	3.33	1.04
1996	0.00	0.00	0.00	0.00	0.00	1.44	13.99	0.00	1.67	0.52
1997	0.00	0.00	0.00	0.00	0.19	2.12	14.67	0.00	2.06	0.66
1998	0.00	0.00	0.00	0.12	0.63	4.26	15.09	0.04	3.45	1.24
1999	0.00	0.00	0.00	0.10	1.01	5.85	20.54	0.04	5.65	2.16
2000	0.00	0.00	0.00	0.38	0.89	5.49	20.04	0.13	6.14	2.41
2001	0.00	0.00	0.16	0.19	1.57	9.36	34.37	0.13	10.58	3.82
2002	0.00	0.00	0.16	1.21	1.54	4.97	30.30	0.49	8.45	3.04
2003	0.00	0.00	0.00	0.00	0.95	2.66	21.53	0.00	5.27	1.70
2004	0.00	0.00	0.00	0.00	0.19	0.65	12.33	0.00	2.23	0.72
Mean	0.00	0.00	0.02	0.17	1.17	6.21	21.33	0.06	3.88	1.26
Median	0.00	0.00	0.00	0.00	0.89	5.49	20.54	0.00	3.45	0.97
Min	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.42	0.09
Max	0.00	0.00	0.26	1.33	5.35	20.78	58.82	0.49	10.58	3.82
StDev	0.00	0.00	0.06	0.32	1.27	4.44	16.78	0.11	2.88	1.05

Exhibit 16 - Annual Issuer-Weighted Default Rates by Alphanumeric Rating, 1983-2004

Cohort Year	Cohort Rating																			All Rated
	Aaa	Aa1	Aa2	Aa3	A1	A2	A3	Baa1	Baa2	Baa3	Ba1	Ba2	Ba3	B1	B2	B3	Caa-C	IG	SG	
1983	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.61	0.00	10.00	17.91	40.00	0.00	3.82	0.97
1984	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.06	1.16	1.61	0.00	5.84	18.75	2.90	0.00	0.10	3.32	0.93
1985	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.63	2.83	4.38	7.69	13.59	0.00	0.00	3.67	1.01
1986	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.82	0.87	1.20	3.44	7.61	16.67	15.79	23.53	0.32	5.64	1.91
1987	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.68	0.95	2.95	4.93	4.30	10.22	20.00	0.00	4.23	1.51
1988	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.56	4.29	7.02	10.96	28.57	0.00	3.59	1.37
1989	0.00	0.00	0.00	1.34	0.00	0.00	0.00	0.00	0.79	1.06	0.79	1.82	4.69	5.73	9.66	18.05	25.00	0.15	5.79	2.27
1990	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.67	2.82	3.91	8.50	22.36	28.93	58.82	0.00	10.08	3.64
1991	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.74	0.00	0.00	1.06	0.00	9.97	5.86	12.74	28.42	36.84	0.07	10.40	3.28
1992	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.73	1.00	1.56	24.54	26.67	0.00	4.85	1.33
1993	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.81	0.00	0.76	3.21	4.96	11.29	28.57	0.00	3.52	0.95
1994	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.59	1.87	3.70	8.05	5.13	0.00	1.95	0.57
1995	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.73	4.31	6.42	4.06	11.57	0.00	3.33	1.04
1996	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.16	0.00	3.28	13.99	0.00	1.67	0.52
1997	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.47	0.00	1.53	7.22	14.67	0.00	2.06	0.66
1998	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.30	0.00	0.00	0.57	1.10	2.07	7.96	5.57	15.09	0.04	3.45	1.24
1999	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.34	0.45	0.00	1.96	3.25	6.65	10.00	19.27	0.04	5.65	2.16
2000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.28	0.00	0.96	0.88	0.61	1.54	3.40	3.94	10.74	19.24	0.13	6.14	2.41
2001	0.00	0.00	0.00	0.00	0.00	0.41	0.00	0.26	0.25	0.00	0.50	1.29	2.90	3.54	10.47	16.32	33.27	0.13	10.58	3.82
2002	0.00	0.00	0.00	0.00	0.00	0.00	0.42	1.23	0.69	1.72	1.52	1.32	1.53	1.88	6.31	6.67	30.29	0.49	8.45	3.04
2003	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.51	0.68	1.41	0.75	2.41	5.98	21.36	0.00	5.27	1.70
2004	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.47	0.00	0.66	1.52	12.33	0.00	2.23	0.72
Mean	0.00	0.00	0.00	0.06	0.00	0.02	0.02	0.11	0.09	0.45	0.68	0.66	2.19	3.34	7.53	11.91	21.92	0.07	4.98	1.68
Median	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.48	0.29	1.64	3.33	6.54	10.48	20.68	0.00	4.03	1.35
Min	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.52	0.00	0.00	1.67	0.52
Max	0.00	0.00	0.00	1.34	0.00	0.41	0.42	1.23	0.79	4.82	3.68	2.82	9.97	8.50	22.36	28.93	58.82	0.49	10.58	3.82
StDev	0.00	0.00	0.00	0.29	0.00	0.09	0.09	0.30	0.23	1.09	0.95	0.81	2.16	2.44	5.88	7.89	13.62	0.12	2.71	1.01

Exhibit 17 - Average Issuer-Weighted Cumulative Default Rates by Whole Letter Rating, 1920-2004

Cohort Rating	Time Horizon (Years)																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	0.00	0.00	0.02	0.09	0.19	0.30	0.41	0.59	0.77	1.01	1.22	1.37	1.57	1.66	1.70	1.80	1.90	1.95	2.07	2.13
Aa	0.06	0.19	0.32	0.49	0.78	1.11	1.48	1.85	2.20	2.57	3.01	3.50	3.98	4.48	4.87	5.13	5.35	5.57	5.87	6.09
A	0.08	0.25	0.54	0.87	1.22	1.58	1.98	2.34	2.76	3.22	3.71	4.21	4.65	5.09	5.56	6.02	6.30	6.60	6.89	7.19
Baa	0.31	0.93	1.69	2.55	3.40	4.28	5.12	5.95	6.83	7.63	8.42	9.22	10.00	10.70	11.32	11.91	12.51	13.04	13.49	13.95
Ba	1.39	3.36	5.48	7.71	9.93	12.01	13.84	15.65	17.25	19.00	20.60	22.16	23.72	25.10	26.31	27.44	28.59	29.70	30.58	31.48
B	4.56	9.97	15.24	19.85	23.80	27.13	30.16	32.62	34.74	36.51	38.24	39.80	41.23	42.67	43.92	45.21	46.15	46.89	47.52	47.79
Caa-C	15.07	24.77	31.82	36.76	40.50	43.63	45.85	47.94	49.89	51.64	53.63	55.61	57.33	59.19	60.94	62.52	63.90	65.34	66.69	67.95
IG	0.15	0.47	0.88	1.34	1.84	2.35	2.88	3.39	3.93	4.47	5.04	5.61	6.15	6.67	7.14	7.57	7.93	8.27	8.60	8.93
SG	3.83	7.78	11.49	14.80	17.73	20.29	22.52	24.53	26.29	28.02	29.67	31.24	32.77	34.19	35.44	36.65	37.76	38.81	39.67	40.45
All Rated	1.48	3.07	4.59	5.99	7.24	8.37	9.38	10.30	11.16	12.01	12.84	13.65	14.43	15.15	15.80	16.41	16.93	17.43	17.88	18.30

Exhibit 18 - Average Issuer-Weighted Cumulative Default Rates by Whole Letter Rating, 1970-2004

Cohort Rating	Time Horizon (Years)																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	0.00	0.00	0.00	0.04	0.12	0.21	0.30	0.41	0.52	0.63	0.76	0.90	1.05	1.13	1.22	1.32	1.42	1.54	1.54	1.54
Aa	0.00	0.00	0.03	0.12	0.20	0.29	0.37	0.47	0.54	0.61	0.69	0.84	1.01	1.25	1.38	1.52	1.73	1.92	2.20	2.44
A	0.02	0.08	0.22	0.36	0.50	0.67	0.85	1.04	1.25	1.48	1.72	1.95	2.20	2.43	2.74	3.12	3.51	3.93	4.41	4.87
Baa	0.19	0.54	0.98	1.55	2.08	2.59	3.12	3.65	4.25	4.89	5.59	6.35	7.12	7.91	8.73	9.48	10.23	10.94	11.56	12.05
Ba	1.22	3.34	5.79	8.27	10.72	12.98	14.81	16.64	18.40	20.11	22.01	24.07	26.11	28.02	29.67	31.53	33.16	34.71	35.92	37.07
B	5.81	12.93	19.51	25.33	30.48	35.10	39.45	42.89	45.89	48.64	50.99	52.85	54.62	56.35	57.72	58.80	59.11	59.11	59.11	59.11
Caa-C	22.43	35.96	46.71	54.19	59.72	64.49	68.06	71.91	74.53	76.77	78.53	78.53	78.53	78.53	78.53	78.53	78.53	78.53	78.53	78.53
IG	0.07	0.21	0.41	0.67	0.92	1.17	1.44	1.70	1.99	2.31	2.64	3.01	3.39	3.77	4.18	4.61	5.04	5.48	5.92	6.31
SG	4.85	9.84	14.43	18.41	21.91	24.95	27.52	29.76	31.75	33.61	35.47	37.27	39.05	40.71	42.13	43.65	44.87	46.00	46.89	47.75
All Rated	1.56	3.15	4.60	5.86	6.94	7.85	8.62	9.30	9.93	10.53	11.14	11.75	12.37	12.95	13.51	14.10	14.65	15.18	15.68	16.13

Exhibit 19 - Average Issuer-Weighted Cumulative Default Rates by Alphanumeric Rating, 1983-2004

Cohort Rating	Time Horizon (Years)																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	0.00	0.00	0.00	0.06	0.18	0.24	0.32	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40
Aa1	0.00	0.00	0.00	0.15	0.15	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.58	0.98	1.48	1.48	1.48	1.48	1.48
Aa2	0.00	0.00	0.04	0.13	0.28	0.34	0.40	0.48	0.57	0.67	0.79	0.93	1.10	1.10	1.10	1.10	1.38	1.75	2.22	2.84
Aa3	0.00	0.00	0.05	0.11	0.18	0.26	0.26	0.26	0.26	0.33	0.41	0.71	1.07	1.21	1.38	1.59	1.82	2.15	2.59	2.59
A1	0.00	0.00	0.19	0.30	0.38	0.47	0.50	0.58	0.67	0.84	1.04	1.21	1.49	1.84	2.25	2.59	2.79	2.79	2.79	2.79
A2	0.03	0.08	0.22	0.47	0.68	0.89	1.05	1.34	1.59	1.69	1.75	1.82	1.90	1.99	2.10	2.35	2.78	3.60	4.39	4.77
A3	0.03	0.21	0.37	0.50	0.65	0.86	1.19	1.38	1.55	1.69	1.85	1.85	1.97	2.10	2.44	3.24	4.12	5.01	6.17	6.73
Baa1	0.17	0.50	0.84	1.14	1.46	1.69	1.92	2.05	2.21	2.31	2.43	2.72	3.08	3.52	4.07	4.41	5.60	6.19	6.19	6.19
Baa2	0.12	0.40	0.81	1.52	2.11	2.74	3.39	3.98	4.62	5.49	6.64	7.91	8.93	9.95	11.48	13.09	13.82	15.43	16.92	17.94
Baa3	0.41	1.07	1.70	2.66	3.60	4.49	5.36	6.15	6.68	7.20	7.53	7.93	9.13	10.29	10.99	10.99	11.45	12.08	13.63	15.70
Ba1	0.66	2.07	3.55	5.23	6.76	8.67	9.70	10.85	11.61	12.38	13.48	14.80	15.59	16.55	16.92	18.27	20.28	22.99	24.73	27.00
Ba2	0.62	2.22	4.48	6.84	8.82	10.11	11.85	13.13	14.20	14.66	16.39	18.23	20.18	22.71	24.28	28.52	29.77	31.75	37.44	37.44
Ba3	2.23	6.10	10.62	15.03	19.14	23.05	26.56	30.00	33.35	36.24	38.25	40.30	42.85	46.05	49.25	50.76	51.64	53.01	53.01	53.01
B1	3.03	8.89	14.81	20.09	25.27	30.29	35.69	39.97	43.98	47.43	50.88	52.62	54.38	56.61	58.42	60.70	60.70	60.70	60.70	60.70
B2	5.93	13.73	20.58	26.58	31.24	34.54	37.39	39.60	42.19	44.48	46.19	47.58	49.27	53.50	56.32	56.32	56.32	56.32	56.32	56.32
B3	10.77	20.43	29.01	36.82	43.55	49.74	54.46	58.40	61.02	62.32	64.06	67.12	70.02	70.02	70.02	70.02	70.02	70.02	70.02	70.02
Caa-C	22.24	35.80	46.75	54.60	60.40	65.15	68.30	72.36	75.38	78.81	81.46	81.46	81.46	81.46	81.46	81.46	81.46	81.46	81.46	81.46
IG	0.08	0.23	0.43	0.71	0.96	1.21	1.43	1.65	1.84	2.03	2.24	2.47	2.77	3.07	3.42	3.81	4.24	4.77	5.39	5.82
SG	5.26	10.84	16.06	20.63	24.54	28.00	31.04	33.63	35.87	37.66	39.45	41.04	42.62	44.50	45.87	47.34	48.27	49.60	50.81	51.68
All Rated	1.79	3.66	5.38	6.89	8.13	9.17	10.04	10.75	11.35	11.83	12.31	12.77	13.27	13.79	14.28	14.80	15.28	15.90	16.56	17.04

Exhibit 20 - Average Issuer-Weighted Default Rates by Alphanumeric Rating, 1998-2004

Cohort Rating	Time Horizon (Years)				
	1	2	3	4	5
Aaa	0.00	0.00	0.00	0.00	0.00
Aa1	0.00	0.00	0.00	0.00	0.00
Aa2	0.00	0.00	0.00	0.00	0.00
Aa3	0.00	0.00	0.00	0.00	0.00
A1	0.00	0.00	0.00	0.00	0.00
A2	0.06	0.14	0.33	0.58	0.85
A3	0.07	0.28	0.50	0.72	0.94
Baa1	0.28	0.68	1.13	1.39	1.52
Baa2	0.19	0.55	1.14	2.00	2.44
Baa3	0.46	1.31	2.13	3.25	4.57
Ba1	0.58	1.91	2.94	3.38	4.21
Ba2	0.66	1.38	3.37	5.43	6.62
Ba3	1.50	4.36	7.84	11.75	14.68
B1	2.30	8.46	14.89	20.51	25.06
B2	5.42	13.62	20.67	27.74	31.77
B3	8.60	19.08	30.01	41.00	49.38
Caa1	14.43	30.86	45.78	56.93	64.95
Caa2	26.53	41.33	52.15	58.80	61.46
Caa3	28.10	41.29	53.13	61.50	66.81
Ca-C	41.40	56.25	66.43	66.43	66.43
IG	0.12	0.34	0.61	0.92	1.19
SG	6.05	12.65	18.80	24.15	27.83
All Rated	2.18	4.56	6.76	8.71	10.04

Annual Rated Defaulted Bond Dollar Volume Totals*, 1970-2004

Cohort Year	IG	SG	All
1970	\$21.30	\$889.17	\$910.47
1971	\$0.00	\$131.84	\$131.84
1972	\$0.00	\$215.08	\$215.08
1973	\$17.37	\$94.30	\$111.67
1974	\$0.00	\$69.36	\$69.36
1975	\$0.00	\$175.60	\$175.60
1976	\$0.00	\$33.97	\$33.97
1977	\$67.97	\$179.49	\$247.46
1978	\$0.00	\$111.88	\$111.88
1979	\$0.00	\$18.36	\$18.36
1980	\$0.00	\$302.17	\$302.17
1981	\$0.00	\$47.46	\$47.46
1982	\$243.10	\$515.12	\$758.22
1983	\$0.00	\$1,160.93	\$1,160.93
1984	\$183.26	\$473.98	\$657.24
1985	\$0.00	\$1,704.81	\$1,704.81
1986	\$178.17	\$3,759.25	\$3,937.42
1987	\$0.00	\$9,131.69	\$9,131.69
1988	\$0.00	\$5,425.33	\$5,425.33
1989	\$1,505.93	\$10,196.25	\$11,702.18
1990	\$0.00	\$20,147.91	\$20,147.91
1991	\$1,348.00	\$15,751.76	\$17,099.76
1992	\$0.00	\$6,274.56	\$6,274.56
1993	\$0.00	\$1,996.76	\$1,996.76
1994	\$0.00	\$2,069.90	\$2,069.90
1995	\$0.00	\$6,724.53	\$6,724.53
1996	\$0.00	\$4,042.29	\$4,042.29
1997	\$0.00	\$5,078.71	\$5,078.71
1998	\$399.00	\$10,271.72	\$10,670.72
1999	\$460.92	\$27,478.06	\$27,938.98
2000	\$4,115.47	\$24,765.85	\$28,881.32
2001	\$21,028.31	\$80,347.17	\$101,375.48
2002	\$55,680.05	\$107,919.11	\$163,599.16
2003	\$0.00	\$34,200.66	\$34,200.66
2004	\$0.00	\$16,019.06	\$16,019.06

* USD Millions

Exhibit 22 - Annual Dollar Volume-Weighted Default Rates

Cohort Year	Aaa	Aa	A	Baa	Ba	B	Caa-C	IG	SG	All
1994	0.00	0.00	0.00	0.00	0.14	1.93	2.80	0.00	1.20	0.24
1995	0.00	0.00	0.00	0.00	0.57	6.26	12.11	0.00	4.28	0.99
1996	0.00	0.00	0.00	0.00	0.00	1.62	22.53	0.00	2.31	0.66
1997	0.00	0.00	0.00	0.00	0.05	3.06	6.70	0.00	1.93	0.50
1998	0.00	0.00	0.00	0.00	0.65	3.02	14.70	0.00	2.98	0.89
1999	0.00	0.00	0.00	0.14	2.22	5.79	20.79	0.03	6.34	1.22
2000	0.00	0.00	0.00	0.65	1.04	5.65	20.46	0.14	5.59	1.30
2001	0.00	0.00	1.18	1.70	1.18	14.25	66.41	0.92	17.33	3.61
2002	0.00	0.00	2.77	3.69	4.07	17.39	84.62	2.32	21.28	5.49
2003	0.00	0.00	0.00	0.00	3.62	2.61	17.08	0.00	5.69	1.00
2004	0.00	0.00	0.00	0.00	0.00	0.20	12.46	0.00	2.48	0.43
Mean	0.00	0.00	0.36	0.56	1.23	5.62	25.51	0.31	6.49	1.48

Exhibit 22 - Annual Dollar Volume-Weighted Default Rates

Cohort										
Median	0.00	0.00	0.00	0.00	0.65	3.06	17.08	0.00	4.28	0.99
Min	0.00	0.00	0.00	0.00	0.00	0.20	2.80	0.00	1.20	0.24
Max	0.00	0.00	2.77	3.69	4.07	17.39	84.62	2.32	21.28	5.49
StDev	0.00	0.00	0.87	1.16	1.46	5.43	25.75	0.72	6.61	1.61

Exhibit 23 - Average Dollar Volume-Weighted Cumulative Default Rates by Whole Letter Rating, 1994-2004

Cohort Rating	Time Horizon (Years)				
	1	2	3	4	5
Aaa	0.00	0.00	0.00	0.00	0.00
Aa	0.00	0.00	0.00	0.00	0.00
A	0.53	1.01	1.41	1.76	2.21
Baa	0.78	1.81	2.81	4.04	4.80
Ba	1.73	4.24	6.93	9.00	10.79
B	5.82	15.02	22.36	27.97	31.77
Caa-C	31.11	47.57	64.26	69.20	70.38
IG	0.47	0.97	1.42	1.89	2.40
SG	7.50	14.81	20.94	25.18	27.98
All Rated	1.82	3.64	5.20	6.49	7.55

Exhibit 24 - Distribution of Annual New Rated Issuers by Whole Letter Rating, 1970-2004

Cohort Year	Percent of Issuers with Initial Rating:							Total Issuers
	Aaa	Aa	A	Baa	Ba	B	Caa-C	
1970	0.00	3.45	37.93	41.38	15.52	1.72	0.00	58
1971	1.56	7.81	28.13	46.88	15.63	0.00	0.00	64
1972	3.85	15.38	19.23	50.00	11.54	0.00	0.00	26
1973	14.29	35.71	42.86	0.00	7.14	0.00	0.00	14
1974	15.22	30.43	43.48	8.70	2.17	0.00	0.00	46
1975	20.69	18.97	55.17	5.17	0.00	0.00	0.00	58
1976	14.71	20.59	47.06	14.71	2.94	0.00	0.00	34
1977	20.59	23.53	17.65	5.88	20.59	11.76	0.00	34
1978	8.00	4.00	14.00	18.00	54.00	2.00	0.00	50
1979	13.33	10.00	26.67	11.67	28.33	8.33	1.67	60
1980	7.84	7.84	18.63	20.59	42.16	2.94	0.00	102
1981	10.81	6.76	28.38	16.22	37.84	0.00	0.00	74

Exhibit 24 - Distribution of Annual New Rated Issuers by Whole Letter Rating, 1970-2004

Cohort	Percent of Issuers with Initial Rating:							Total
1982	17.02	17.73	28.37	14.89	14.89	7.09	0.00	141
1983	12.04	12.04	21.30	12.04	26.85	15.74	0.00	108
1984	8.05	19.92	23.73	11.44	25.00	10.59	1.27	236
1985	10.25	14.84	22.26	9.89	31.10	11.31	0.35	283
1986	7.74	10.03	13.18	8.88	38.40	21.49	0.29	349
1987	8.19	13.99	9.90	9.22	33.45	25.26	0.00	293
1988	10.17	22.03	16.53	6.78	24.58	19.49	0.42	236
1989	9.59	26.94	20.30	7.01	21.03	15.13	0.00	271
1990	11.26	38.41	20.53	14.57	7.95	7.28	0.00	151
1991	9.68	30.41	30.41	11.98	12.90	4.61	0.00	217
1992	6.92	20.44	20.75	16.35	20.44	14.15	0.94	318
1993	2.99	11.78	20.16	16.17	23.15	21.96	3.79	501
1994	2.14	14.78	22.06	11.99	11.78	31.91	5.35	467
1995	2.70	11.01	21.12	19.78	12.58	29.44	3.37	445
1996	2.14	7.14	18.21	20.00	19.82	28.57	4.11	560
1997	0.86	5.32	13.81	18.99	19.14	36.12	5.76	695
1998	2.12	8.06	14.29	14.71	15.13	36.63	9.05	707
1999	2.20	9.36	15.78	19.63	16.51	31.38	5.14	545
2000	1.26	13.84	27.99	22.33	8.81	22.33	3.46	318
2001	2.86	14.84	25.00	28.39	8.85	17.97	2.08	384
2002	0.94	14.15	24.53	27.99	7.86	22.96	1.57	318
2003	2.54	13.23	19.34	18.58	10.94	31.30	4.07	393
2004	0.55	15.53	28.10	9.61	6.84	23.84	15.53	541

Exhibit 25 - Rating Migration for Defaulters from Five Years Prior to Default, 1983-2004

2004 Cohort		Years Prior to Default					
Statistic		0	1	2	3	4	5
Mean		Caa3	Caa2/Caa1	Caa1/B3	B3	B3/B2	B2/B1
Median		Caa3	Caa2	Caa1	B3	B3	B2
High		B1	Ba3	Baa2	Baa2	Baa2	A1
Low		C	Ca	Ca	Caa3	Caa3	Caa3
Total Defaults		34					
1983-2003		Years Prior to Default					
Statistic		0	1	2	3	4	5
Mean		Caa2/Caa1	B2	B1	B1/Ba3	Ba3	Ba3/Ba2
Median		Caa1	B2	B1	B1	B1	Ba3
High		Baa2	Aa3	Aa2	Aa2	Aaa	Aaa
Low		C	C	C	C	C	Ca
Total Defaults		1,187					

Exhibit 26 - Cumulative Issuer-Weighted Default Rates by Annual Cohort, 1970-2004

Cohort Year	Cohort Rating	Gross Denominator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1970	Aaa	39	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.70	2.70	2.70
	Aa	76	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.48	1.48	1.48	2.99	2.99	2.99	2.99
	A	245	0.00	0.00	0.00	0.00	0.00	0.43	0.43	0.43	0.43	0.43	0.90	0.90	0.90	0.90	0.90	0.90	1.41	1.41	2.54	2.54
	Baa	370	0.27	0.27	0.27	1.14	1.43	1.43	1.75	2.40	3.07	3.07	3.44	3.44	4.62	5.03	5.46	6.33	7.73	8.75	9.86	10.45
	Ba	247	4.12	4.54	4.98	5.44	6.38	7.37	7.88	8.44	9.60	9.60	9.60	11.06	13.41	14.25	14.25	16.16	20.26	22.55	22.55	23.95
	B	39	20.78	23.38	25.98	25.98	25.98	25.98	25.98	25.98	25.98	25.98	25.98	25.98	33.38	33.38	33.38	33.38	33.38	33.38	33.38	33.38
	Caa-C	15	53.33	60.00	80.00	88.00	88.00	88.00	88.00	100.00												
	IG	730	0.14	0.14	0.14	0.57	0.72	0.87	1.02	1.33	1.65	1.65	1.98	1.98	2.51	2.88	3.06	3.44	4.41	5.03	5.90	6.12
	SG	301	8.78	9.82	11.61	12.36	13.12	13.93	14.35	15.24	16.18	16.18	16.18	17.34	20.44	21.10	21.10	22.60	25.82	27.63	27.63	28.74
	All Rated	1,031	2.64	2.94	3.45	3.97	4.29	4.61	4.83	5.29	5.77	5.77	6.02	6.29	7.38	7.80	7.95	8.54	9.93	10.76	11.46	11.83
1971	Aaa	39	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.78	2.78	2.78	2.78
	Aa	73	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.82
	A	273	0.00	0.00	0.00	0.00	0.38	0.38	0.38	0.38	0.38	0.79	0.79	1.22	1.64	1.64	1.64	2.08	2.08	3.05	3.05	3.05
	Baa	397	0.00	0.00	0.80	1.07	1.07	1.35	1.96	2.58	2.58	2.92	2.92	4.04	4.43	4.83	5.65	6.98	7.92	8.96	9.50	10.67
	Ba	240	0.42	0.86	1.31	2.25	3.71	4.22	4.76	5.92	5.92	5.92	7.35	9.64	10.46	10.46	12.31	17.29	19.56	19.56	20.93	20.93
	B	26	3.85	7.70	7.70	7.70	7.70	7.70	7.70	7.70	7.70	7.70	7.70	20.00	20.00	20.00	20.00	20.00	20.00	20.00	20.00	20.00
	Caa-C	8	13.33	53.33	68.88	68.88	68.88	68.88	89.63	89.63	89.63	89.63	89.63	89.63	89.63	89.63	89.63	89.63	89.63	89.63	89.63	89.63
	IG	782	0.00	0.00	0.40	0.53	0.67	0.81	1.10	1.40	1.40	1.71	1.71	2.37	2.71	2.88	3.23	3.96	4.53	5.34	5.55	6.20
	SG	274	1.10	3.01	3.80	4.62	5.90	6.35	7.30	8.31	8.31	8.31	9.55	12.86	13.57	13.57	15.16	19.45	21.38	21.38	22.57	22.57
	All Rated	1,056	0.29	0.77	1.27	1.57	1.99	2.20	2.65	3.10	3.10	3.35	3.60	4.79	5.20	5.34	5.91	7.25	8.04	8.72	9.07	9.63
1972	Aaa	40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.70	2.70	2.70	2.70	2.70
	Aa	76	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.72	1.72
	A	295	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.39	0.79	0.79	0.79	1.20	1.20	2.08	2.08	2.57	3.07
	Baa	430	0.00	0.73	0.98	1.24	1.50	2.05	2.63	2.63	3.26	3.26	3.96	4.33	4.70	5.47	6.68	7.56	8.51	9.53	11.15	13.42
	Ba	236	0.00	0.45	1.37	2.81	3.31	3.84	4.97	4.97	4.97	6.34	9.28	10.06	10.89	13.53	18.23	20.34	20.34	21.59	22.90	29.79
	B	28	7.14	7.14	7.14	7.14	7.14	7.14	7.14	7.14	7.14	7.14	17.75	17.75	17.75	17.75	17.75	17.75	17.75	17.75	17.75	17.75
	Caa-C	8	40.00	57.14	57.14	57.14	57.14	78.57	78.57	78.57	78.57	78.57	78.57	78.57	78.57							
	IG	841	0.00	0.37	0.49	0.62	0.75	1.02	1.29	1.29	1.58	1.58	2.04	2.35	2.51	2.84	3.51	4.05	4.80	5.19	6.21	7.27
	SG	272	1.88	2.65	3.45	4.71	5.15	6.07	7.04	7.04	7.04	8.23	12.00	12.67	13.38	15.65	19.71	21.51	21.51	22.60	23.76	29.86
	All Rated	1,113	0.46	0.92	1.21	1.60	1.80	2.21	2.64	2.64	2.87	3.11	4.22	4.60	4.86	5.53	6.78	7.51	8.14	8.63	9.67	11.45

Exhibit 26 - Cumulative Issuer-Weighted Default Rates by Annual Cohort, 1970-2004

Cohort Year	Cohort Rating	Gross Denominator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1973	Aaa	40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.67	2.67	2.67	2.67	2.67	2.67
	Aa	80	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.59	1.59	1.59
	A	300	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.38	0.77	0.77	0.77	1.17	1.17	2.04	2.04	2.51	2.99	3.52
	Baa	444	0.46	0.69	1.17	1.41	1.93	2.46	2.46	3.05	3.05	3.70	4.37	5.06	6.13	7.26	8.07	8.95	9.90	11.91	14.03	14.03
	Ba	213	0.00	0.97	1.99	2.53	3.09	4.28	4.28	4.28	5.70	9.49	10.29	10.29	12.06	16.82	20.03	20.03	21.29	22.61	30.83	32.30
	B	27	3.77	3.77	3.77	3.77	3.77	3.77	3.77	3.77	3.77	15.09	15.09	15.09	15.09	15.09	15.09	15.09	15.09	15.09	15.09	15.09
	Caa-C	5	44.44	44.44	44.44	44.44	66.66	66.66	66.66	66.66	66.66	66.66	66.66	66.66	66.66	66.66	66.66	66.66	66.66	66.66	66.66	66.66
	IG	864	0.23	0.35	0.59	0.72	0.97	1.23	1.23	1.50	1.50	1.94	2.39	2.69	3.16	3.80	4.30	5.02	5.39	6.56	7.57	7.78
	SG	245	1.24	2.09	2.98	3.45	4.44	5.47	5.47	5.47	6.70	11.27	11.96	11.96	13.51	17.69	20.49	20.49	21.61	22.81	30.28	31.61
	All Rated	1,109	0.45	0.73	1.11	1.31	1.71	2.12	2.12	2.35	2.58	3.77	4.26	4.52	5.16	6.37	7.23	7.83	8.31	9.48	11.38	11.75
1974	Aaa	44	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.41	2.41	2.41	2.41	2.41	2.41	2.41
	Aa	87	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.27	2.79	2.79	2.79	2.79
	A	297	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.38	0.77	0.77	0.77	1.17	1.17	2.03	2.03	2.49	2.96	3.48	3.48
	Baa	437	0.00	0.47	0.72	1.23	1.76	1.76	2.34	2.34	2.97	3.63	4.32	5.38	5.75	6.56	7.44	8.38	9.88	11.99	11.99	11.99
	Ba	201	0.00	1.06	1.62	2.21	3.47	3.47	3.47	4.22	8.17	9.01	9.01	10.84	17.70	21.01	21.01	22.32	25.04	33.45	34.96	36.67
	B	31	10.00	10.00	10.00	10.00	10.00	10.00	10.00	14.39	23.65	23.65	23.65	23.65	23.65	23.65	23.65	23.65	23.65	23.65	23.65	23.65
	Caa-C	2	0.00	0.00	0.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00
	IG	865	0.00	0.24	0.36	0.60	0.86	0.86	1.13	1.13	1.70	2.14	2.44	2.90	3.21	3.71	4.41	4.78	5.73	6.71	6.92	6.92
	SG	234	1.31	2.22	2.70	3.72	4.78	4.78	4.78	6.05	10.73	11.45	11.45	13.04	19.04	21.91	21.91	23.07	25.55	33.32	34.71	36.26
	All Rated	1,099	0.28	0.65	0.85	1.24	1.65	1.65	1.88	2.10	3.40	3.89	4.14	4.78	5.98	6.82	7.42	7.89	9.04	10.91	11.27	11.47
1975	Aaa	50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.08	2.08	2.08	2.08	2.08	2.08	2.08	2.08
	Aa	102	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.05	1.05	1.05	1.05	1.05	1.05	2.25	2.25	3.49	3.49	3.49	3.49	3.49
	A	308	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.37	0.37	0.37	0.76	0.76	1.59	1.59	2.04	2.98	3.48	4.02	4.02
	Baa	425	0.00	0.00	0.26	0.80	0.80	1.38	1.38	2.34	3.00	3.68	4.74	5.11	5.91	6.34	7.28	9.26	11.32	11.32	11.32	11.32
	Ba	201	1.02	2.10	3.23	3.83	3.83	3.83	4.56	8.35	9.15	9.15	10.90	17.46	20.64	21.81	23.06	24.36	31.05	32.49	34.07	34.07
	B	34	5.97	5.97	5.97	9.27	9.27	9.27	13.14	21.21	21.21	21.21	21.21	21.21	21.21	21.21	21.21	31.06	42.55	42.55	42.55	42.55
	Caa-C	2	0.00	0.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00
	IG	885	0.00	0.00	0.12	0.36	0.36	0.62	0.62	1.17	1.59	1.87	2.31	2.61	3.08	3.75	4.09	5.17	6.29	6.48	6.70	6.70
	SG	237	1.73	2.64	4.07	5.07	5.07	5.07	6.26	10.70	11.37	11.37	12.88	18.55	21.27	22.26	23.33	25.62	32.76	34.03	35.42	35.42
	All Rated	1,122	0.36	0.55	0.93	1.32	1.32	1.53	1.75	2.99	3.45	3.69	4.29	5.43	6.23	6.94	7.38	8.62	10.53	10.87	11.24	11.24

Exhibit 26 - Cumulative Issuer-Weighted Default Rates by Annual Cohort, 1970-2004

Cohort Year	Cohort Rating	Gross Denominator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1976	Aaa	65	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.69	1.69	1.69	1.69	1.69	1.69	1.69	1.69	1.69
	Aa	108	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00	2.15	2.15	3.34	3.34	3.34	3.34	3.34	3.34
	A	356	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.64	0.64	0.64	1.32	1.32	2.43	2.43	2.83	4.07	4.07	4.56	4.56	5.61
	Baa	387	0.00	0.27	0.56	0.56	0.87	0.87	2.22	2.92	3.64	4.77	5.16	5.99	5.99	6.97	9.05	10.68	11.24	11.24	11.24	11.24
	Ba	203	1.01	2.08	3.20	3.20	3.81	4.48	7.31	8.05	8.05	9.68	15.75	18.69	19.78	20.94	22.14	29.62	30.95	32.42	32.42	32.42
	B	30	0.00	0.00	3.64	3.64	3.64	8.12	17.55	17.55	17.55	17.55	17.55	17.55	17.55	17.55	30.23	45.73	45.73	45.73	45.73	45.73
	Caa-C	2	0.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00
	IG	916	0.00	0.11	0.23	0.23	0.35	0.35	0.99	1.52	1.78	2.20	2.62	3.07	3.70	4.03	5.05	6.11	6.30	6.50	6.50	6.94
	SG	235	0.87	2.25	3.68	3.68	4.22	5.37	9.02	9.67	9.67	11.12	16.55	19.16	20.12	21.15	23.33	31.34	32.57	33.92	33.92	33.92
	All Rated	1,151	0.18	0.54	0.91	0.91	1.11	1.31	2.48	3.02	3.25	3.82	5.02	5.78	6.46	6.88	8.05	10.02	10.34	10.70	10.70	11.08
1977	Aaa	65	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.71	1.71	1.71	1.71	1.71	1.71	1.71	1.71	1.71	1.71
	Aa	115	0.00	0.00	0.00	0.00	0.00	0.93	0.93	0.93	0.93	0.93	0.93	2.01	2.01	3.13	3.13	3.13	3.13	3.13	3.13	3.13
	A	374	0.00	0.00	0.00	0.00	0.00	0.00	0.60	0.60	0.60	1.25	1.25	3.01	3.01	3.77	4.95	4.95	5.42	5.42	6.42	6.42
	Baa	372	0.28	0.57	0.57	0.57	0.57	1.93	2.64	3.37	4.51	4.91	5.76	5.76	6.74	8.32	9.95	10.52	10.52	10.52	10.52	10.52
	Ba	197	0.52	1.61	1.61	2.22	2.88	5.64	6.36	6.36	7.95	13.83	16.66	17.69	18.78	19.94	27.16	28.46	29.89	29.89	29.89	29.89
	B	31	3.28	6.73	6.73	10.98	15.54	25.78	25.78	25.78	25.78	25.78	25.78	25.78	25.78	39.27	56.62	56.62	56.62	56.62	56.62	56.62
	Caa-C	2	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00
	IG	926	0.11	0.22	0.22	0.22	0.22	0.85	1.36	1.62	2.02	2.44	2.87	3.79	4.11	5.11	6.13	6.31	6.51	6.51	6.95	6.95
	SG	230	1.34	2.75	2.75	3.80	4.93	8.52	9.16	9.16	10.58	15.89	18.42	19.35	20.34	22.43	30.19	31.39	32.71	32.71	32.71	32.71
	All Rated	1,156	0.35	0.71	0.71	0.91	1.11	2.24	2.78	2.99	3.56	4.73	5.47	6.39	6.80	7.94	9.85	10.17	10.51	10.51	10.88	10.88
1978	Aaa	70	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.56	1.56	1.56	3.14	3.14	3.14	3.14	3.14	3.14	3.14	3.14
	Aa	128	0.00	0.00	0.00	0.00	0.84	0.84	0.84	0.84	0.84	0.84	1.78	1.78	1.78	1.78	1.78	1.78	1.78	1.78	1.78	1.78
	A	363	0.00	0.00	0.00	0.00	0.00	0.61	0.61	0.61	1.27	1.27	2.73	2.73	3.91	4.74	4.74	5.23	5.23	6.27	6.27	6.27
	Baa	357	0.00	0.00	0.00	0.00	1.32	1.67	2.37	3.12	3.50	4.74	5.19	6.13	7.65	9.74	10.29	10.29	10.29	10.29	10.29	10.29
	Ba	189	1.08	1.08	1.08	1.73	4.50	5.96	5.96	9.12	14.95	16.81	17.84	18.93	21.23	28.45	29.75	32.61	32.61	32.61	32.61	32.61
	B	38	5.41	5.41	11.72	15.05	22.28	22.28	26.71	26.71	32.35	38.50	38.50	38.50	47.29	60.47	60.47	60.47	60.47	60.47	60.47	60.47
	Caa-C	3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00
	IG	918	0.00	0.00	0.00	0.00	0.61	0.98	1.24	1.50	1.91	2.47	3.37	3.68	4.82	5.82	5.99	6.19	6.19	6.61	6.61	6.61
	SG	230	1.78	1.78	2.78	3.86	7.28	8.49	9.13	11.80	18.21	20.58	21.44	22.37	25.34	32.75	33.91	36.46	36.46	36.46	36.46	36.46
	All Rated	1,148	0.35	0.35	0.54	0.74	1.84	2.36	2.67	3.33	4.69	5.53	6.42	6.82	8.20	10.06	10.37	10.87	10.87	11.23	11.23	11.23

Exhibit 26 - Cumulative Issuer-Weighted Default Rates by Annual Cohort, 1970-2004

Cohort Year	Cohort Rating	Gross Denominator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1979	Aaa	73	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.47	1.47	1.47	2.97	2.97	2.97	2.97	2.97	2.97	2.97	2.97	2.97
	Aa	129	0.00	0.00	0.00	0.82	0.82	0.82	0.82	0.82	0.82	1.74	1.74	1.74	1.74	1.74	1.74	1.74	1.74	1.74	1.74	1.74
	A	360	0.00	0.00	0.00	0.00	0.60	0.60	0.60	1.25	1.25	2.69	2.69	3.47	4.28	4.28	4.76	4.76	5.78	5.78	5.78	5.78
	Baa	351	0.00	0.30	0.30	1.61	1.94	2.29	3.02	3.02	4.23	4.67	5.62	8.15	10.26	10.80	10.80	10.80	10.80	10.80	10.80	10.80
	Ba	210	0.49	0.49	1.05	3.39	5.85	9.09	11.79	18.20	19.75	20.60	21.52	24.40	31.56	32.65	35.01	35.01	35.01	35.01	35.01	35.01
	B	35	0.00	6.06	9.24	16.10	16.10	20.19	24.63	35.02	40.93	40.93	40.93	49.37	49.37	49.37	49.37	49.37	49.37	49.37	49.37	49.37
	Caa-C	3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00
	IG	913	0.00	0.11	0.11	0.71	1.08	1.20	1.46	1.72	2.28	3.16	3.47	4.74	5.72	5.90	6.09	6.09	6.51	6.51	6.51	6.51
	SG	248	0.42	1.29	2.23	5.18	7.26	10.58	13.48	20.87	22.90	23.63	24.43	27.80	34.20	35.20	37.36	37.36	37.36	37.36	37.36	37.36
	All Rated	1,161	0.09	0.36	0.55	1.61	2.30	3.02	3.75	5.28	6.09	6.95	7.33	8.94	10.74	11.03	11.52	11.52	11.87	11.87	11.87	11.87
1980	Aaa	85	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.27	1.27	1.27	2.57	2.57	2.57	2.57	2.57	2.57	2.57	2.57	2.57	2.57
	Aa	125	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.93	1.87	1.87	1.87	1.87	1.87	1.87	1.87	1.87	1.87	1.87	1.87	1.87
	A	369	0.00	0.00	0.28	0.86	0.86	0.86	1.78	2.10	3.14	3.14	3.89	4.67	4.67	5.12	5.12	6.10	6.10	6.10	6.10	6.64
	Baa	351	0.00	0.00	0.96	1.30	1.64	2.70	2.70	3.49	4.34	5.70	8.10	10.10	11.14	11.14	11.14	11.14	11.14	11.14	11.14	11.14
	Ba	210	0.00	0.53	3.83	4.99	8.64	11.81	17.85	20.05	20.86	23.55	26.38	34.50	36.74	39.15	39.15	39.15	39.15	40.55	40.55	40.55
	B	41	4.94	7.54	15.83	21.74	28.12	31.63	44.06	48.72	48.72	48.72	60.11	68.09	68.09	68.09	68.09	68.09	68.09	68.09	68.09	68.09
	Caa-C	6	33.33	33.33	33.33	33.33	33.33	33.33	55.55	55.55	55.55	55.55	55.55	55.55	55.55	55.55	55.55	55.55	55.55	55.55	55.55	55.55
	IG	930	0.00	0.00	0.46	0.81	0.93	1.31	1.69	2.36	3.20	3.64	4.86	5.81	6.14	6.32	6.32	6.72	6.72	6.72	6.72	6.95
	SG	257	1.61	2.46	6.49	8.37	12.36	15.50	22.78	25.23	25.91	28.12	32.05	39.84	41.78	43.88	43.88	43.88	43.88	45.08	45.08	45.08
	All Rated	1,187	0.34	0.52	1.71	2.37	3.25	4.15	5.81	6.79	7.61	8.34	9.99	11.96	12.52	12.99	12.99	13.32	13.32	13.50	13.50	13.69
1981	Aaa	89	0.00	0.00	0.00	0.00	0.00	0.00	1.27	1.27	1.27	2.58	2.58	2.58	2.58	2.58	2.58	2.58	2.58	2.58	2.58	2.58
	Aa	130	0.00	0.00	0.00	0.00	0.00	0.00	0.86	2.59	2.59	2.59	2.59	2.59	2.59	2.59	2.59	2.59	2.59	2.59	2.59	3.93
	A	376	0.00	0.27	0.27	0.27	0.27	1.18	1.50	2.19	2.19	2.94	3.71	3.71	4.18	4.18	5.16	5.16	5.16	5.16	5.70	6.25
	Baa	352	0.00	0.61	1.88	2.55	3.58	3.58	3.97	4.78	6.09	8.40	9.83	10.81	10.81	10.81	10.81	10.81	10.81	10.81	10.81	11.56
	Ba	244	0.00	3.59	5.00	7.98	11.62	18.29	20.75	21.44	24.45	28.47	36.51	38.55	40.74	40.74	40.74	40.74	42.04	43.39	43.39	43.39
	B	46	4.49	11.57	16.55	24.49	27.34	40.26	40.26	40.26	40.26	50.22	56.44	56.44	56.44	56.44	56.44	56.44	56.44	56.44	56.44	56.44
	Caa-C	6	0.00	0.00	0.00	0.00	0.00	20.00	40.00	40.00	40.00	40.00	40.00	40.00	40.00	40.00	40.00	40.00	40.00	40.00	40.00	40.00
	IG	947	0.00	0.33	0.78	1.02	1.37	1.75	2.27	3.09	3.52	4.71	5.47	5.79	5.97	5.97	6.36	6.36	6.36	6.36	6.58	7.25
	SG	296	0.70	4.75	6.69	10.38	13.82	21.68	24.22	24.78	27.25	31.94	39.50	41.23	43.11	43.11	43.11	43.11	44.22	45.36	45.36	45.36
	All Rated	1,243	0.16	1.35	2.14	3.13	4.16	6.11	7.04	7.81	8.62	10.42	12.31	12.84	13.28	13.28	13.60	13.60	13.78	13.95	14.14	14.69

Exhibit 26 - Cumulative Issuer-Weighted Default Rates by Annual Cohort, 1970-2004

Cohort Year	Cohort Rating	Gross Denominator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1982	Aaa	91	0.00	0.00	0.00	0.00	0.00	1.24	1.24	1.24	2.54	2.54	2.54	2.54	2.54	2.54	2.54	2.54	2.54	2.54	2.54	2.54
	Aa	142	0.00	0.00	0.00	0.00	0.00	0.78	2.36	2.36	2.36	2.36	2.36	2.36	2.36	3.49	3.49	3.49	3.49	3.49	4.74	4.74
	A	387	0.26	0.26	0.26	0.26	1.14	1.14	1.80	1.80	2.88	3.64	3.64	4.08	4.08	4.08	4.08	4.08	4.08	4.60	5.12	8.35
	Baa	337	0.31	0.31	1.33	2.38	2.75	3.53	4.37	5.71	8.06	9.52	10.53	10.53	10.53	11.15	11.15	11.15	11.15	11.15	11.91	11.91
	Ba	261	2.72	5.19	7.83	11.48	18.28	20.43	21.02	23.65	27.97	32.86	34.72	36.73	36.73	36.73	36.73	38.02	39.40	39.40	39.40	42.36
	B	42	2.41	9.91	15.14	17.92	30.31	30.31	30.31	30.31	35.67	60.18	60.18	60.18	60.18	60.18	60.18	60.18	60.18	60.18	60.18	60.18
	Caa-C	12	25.00	50.00	50.00	50.00	59.09	68.18	68.18	68.18	68.18	68.18	68.18	68.18	68.18	68.18	68.18	68.18	68.18	68.18	68.18	68.18
	IG	957	0.21	0.21	0.55	0.90	1.38	1.89	2.70	3.12	4.43	5.18	5.50	5.68	5.68	6.06	6.06	6.06	6.06	6.28	6.94	8.28
	SG	315	3.54	7.62	10.51	13.89	21.53	23.75	24.24	26.43	30.65	37.48	39.04	40.74	40.74	40.74	40.74	41.80	42.93	42.93	42.93	45.39
	All Rated	1,272	1.04	2.03	2.98	4.04	6.18	7.06	7.80	8.57	10.44	12.26	12.78	13.21	13.21	13.52	13.52	13.69	13.87	14.04	14.59	16.07
1983	Aaa	91	0.00	0.00	0.00	0.00	2.55	2.55	2.55	3.93	3.93	3.93	3.93	3.93	3.93	3.93	3.93	3.93	3.93	3.93	3.93	3.93
	Aa	218	0.00	0.00	0.00	0.00	0.50	2.04	2.04	2.04	2.04	2.04	2.04	2.04	2.77	2.77	2.77	2.77	2.77	2.77	3.57	4.40
	A	432	0.00	0.00	0.00	0.26	0.26	0.83	0.83	1.76	2.72	3.40	3.77	3.77	3.77	3.77	3.77	3.77	3.77	4.24	6.16	6.66
	Baa	274	0.00	1.16	1.57	3.28	3.75	4.27	5.37	6.53	7.74	7.74	7.74	7.74	7.74	7.74	7.74	8.69	10.63	11.62	13.66	
	Ba	230	0.91	2.40	5.05	12.69	14.74	17.88	21.50	26.59	32.19	32.19	33.62	33.62	35.20	35.20	37.03	38.91	38.91	40.92	45.14	49.52
	B	114	6.31	11.02	17.86	25.14	28.42	29.60	32.33	40.29	50.24	55.22	58.02	58.02	58.02	58.02	58.02	58.02	58.02	58.02	58.02	58.02
	Caa-C	11	40.00	60.00	60.00	60.00	70.00	70.00	70.00	70.00	70.00	70.00	70.00	70.00	70.00	70.00	70.00	70.00	70.00	70.00	70.00	70.00
	IG	1,015	0.00	0.31	0.41	0.96	1.42	2.16	2.42	3.22	3.91	4.20	4.36	4.36	4.54	4.54	4.54	4.54	4.74	5.35	6.59	7.44
	SG	355	3.82	6.96	10.96	18.25	21.10	23.39	26.55	32.55	39.40	40.99	42.77	42.77	43.75	43.75	44.90	46.11	46.11	47.39	50.09	52.90
	All Rated	1,370	0.97	1.97	3.01	5.12	6.11	7.17	7.99	9.71	11.41	11.89	12.29	12.29	12.59	12.59	12.75	12.91	13.08	13.76	15.15	16.22
1984	Aaa	0.00	0.00	0.00	1.54	1.54	1.54	3.26	3.26	3.26	3.26	3.26	3.26	3.26	3.26	3.26	3.26	3.26	3.26	3.26	3.26	0.00
	Aa	0.00	0.00	0.00	0.90	1.84	1.84	1.84	1.84	1.84	1.84	1.84	2.49	2.49	2.49	2.49	2.49	2.49	3.21	3.96	3.96	0.00
	A	0.00	0.22	0.45	0.70	1.47	1.74	2.57	3.44	4.06	4.06	4.06	4.06	4.06	4.06	4.06	4.06	4.48	6.21	6.66	7.15	0.00
	Baa	0.36	0.36	0.77	1.22	1.73	2.80	3.94	5.72	5.72	6.42	6.42	6.42	6.42	6.42	6.42	7.35	9.26	10.25	12.30	13.38	0.36
	Ba	0.83	3.93	12.60	14.78	18.51	22.04	27.55	34.41	35.37	36.45	36.45	37.67	37.67	39.14	39.14	39.14	40.78	44.16	47.65	47.65	0.83
	B	6.72	12.76	20.11	23.98	27.13	32.07	42.30	49.51	51.75	56.83	56.83	56.83	56.83	56.83	60.94	60.94	60.94	60.94	60.94	60.94	6.72
	Caa-C	2	100.00																			
	IG	2,144	0.10	0.19	0.40	0.95	1.65	2.01	2.77	3.56	3.84	3.99	3.99	4.16	4.16	4.16	4.16	4.35	4.94	6.13	6.95	7.38
	SG	646	3.34	7.42	15.59	18.37	21.88	25.91	33.04	39.93	41.26	43.49	43.49	44.34	44.34	45.35	46.46	46.46	47.64	50.07	52.60	52.60
	All Rated	2,790	0.93	2.03	4.21	5.29	6.63	7.76	9.77	11.67	12.12	12.62	12.62	12.90	12.90	13.05	13.20	13.36	14.00	15.32	16.34	16.70

Exhibit 26 - Cumulative Issuer-Weighted Default Rates by Annual Cohort, 1970-2004

Cohort Year	Cohort Rating	Gross Denominator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1985	Aaa	75	0.00	0.00	0.00	0.00	0.00	1.75	1.75	1.75	1.75	1.75	1.75	1.75	1.75	1.75	1.75	1.75	1.75	1.75	1.75	1.75
	Aa	296	0.00	0.00	0.00	0.79	0.79	0.79	0.79	0.79	0.79	0.79	1.40	1.40	1.40	1.40	1.40	1.40	2.07	2.76	2.76	2.76
	A	524	0.00	0.21	1.33	2.27	2.52	3.56	4.38	4.67	4.67	4.67	4.67	4.67	4.67	4.67	4.67	5.07	6.29	6.72	7.16	7.16
	Baa	278	0.00	1.20	1.20	1.71	2.79	3.36	5.14	5.77	6.47	6.47	6.47	6.47	6.47	6.47	8.47	10.55	11.63	13.88	15.06	15.06
	Ba	294	1.40	6.69	9.28	12.75	18.29	23.88	30.71	32.21	33.05	33.05	34.99	34.99	36.11	36.11	36.11	37.46	41.72	44.64	44.64	44.64
	B	148	8.22	17.65	23.84	27.26	31.51	43.36	50.13	52.39	57.68	57.68	57.68	57.68	57.68	61.92	61.92	61.92	61.92	61.92	61.92	61.92
	Caa-C	5	0.00	0.00	0.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	62.50	62.50	62.50	62.50	62.50	62.50	62.50	62.50	62.50	62.50
	IG	1,173	0.00	0.37	0.87	1.62	1.96	2.67	3.40	3.67	3.81	3.81	3.97	3.97	3.97	3.97	4.34	4.91	5.87	6.66	7.08	7.08
	SG	447	3.67	10.30	14.14	17.88	22.90	30.34	36.95	38.58	40.41	40.41	42.51	42.51	43.32	44.21	44.21	45.21	48.37	50.54	50.54	50.54
	All Rated	1,620	1.01	3.09	4.47	5.99	7.44	9.63	11.56	12.08	12.54	12.54	13.06	13.06	13.20	13.34	13.64	14.25	15.51	16.48	16.82	16.82
1986	Aaa	97	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Aa	293	0.00	0.00	0.79	0.79	1.23	1.23	1.23	1.23	1.23	1.82	1.82	1.82	1.82	1.82	1.82	2.50	3.19	3.19	3.19	3.19
	A	579	0.00	0.19	0.79	1.21	1.86	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.98	4.02	4.75	5.13	5.13	5.13
	Baa	311	1.33	1.33	3.01	3.88	5.26	6.71	7.76	8.35	8.35	8.35	8.35	8.35	8.35	10.04	10.92	11.83	13.69	13.69	13.69	13.69
	Ba	359	2.03	5.92	8.51	13.53	20.13	28.09	29.92	32.66	33.41	34.20	34.20	36.09	36.09	36.09	38.44	43.41	44.71	46.05	46.05	46.05
	B	185	11.73	17.70	21.70	25.73	35.50	45.06	49.51	53.00	53.00	55.41	55.41	55.41	58.59	62.35	62.35	62.35	62.35	62.35	62.35	62.35
	Caa-C	10	23.53	23.53	36.28	36.28	36.28	36.28	36.28	36.28	36.28	61.77	61.77	61.77	61.77	61.77	61.77	61.77	61.77	61.77	61.77	61.77
	IG	1,280	0.32	0.41	1.24	1.62	2.33	2.86	3.09	3.21	3.21	3.35	3.35	3.35	3.35	3.67	4.16	5.00	5.86	6.04	6.04	6.04
	SG	554	5.64	10.19	13.47	18.06	25.51	33.73	36.24	39.09	39.62	41.33	41.33	42.67	43.42	44.24	45.94	49.54	50.49	51.46	51.46	51.46
	All Rated	1,834	1.91	3.31	4.83	6.33	8.68	10.94	11.64	12.32	12.42	12.86	12.86	13.10	13.22	13.61	14.28	15.49	16.34	16.63	16.63	16.63
1987	Aaa	122	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Aa	308	0.00	0.00	0.00	0.40	0.40	0.40	0.40	0.40	0.95	0.95	0.95	0.95	0.95	0.95	1.58	2.23	2.23	2.23	2.23	2.23
	A	555	0.00	0.00	0.40	1.24	1.67	1.67	1.67	1.67	1.67	1.67	1.67	1.67	1.67	2.00	2.65	3.34	4.07	4.07	4.07	4.07
	Baa	339	0.00	1.05	1.42	2.98	4.62	5.93	6.89	6.89	6.89	6.89	7.54	7.54	8.94	9.66	11.89	12.65	12.65	12.65	12.65	12.65
	Ba	465	2.71	4.47	9.26	16.27	23.97	27.38	30.86	31.97	33.20	33.88	35.46	36.35	36.35	38.42	40.60	44.09	45.35	45.35	45.35	45.35
	B	265	6.23	13.49	20.35	31.89	43.12	46.57	48.63	48.63	50.00	50.00	50.00	51.96	54.14	58.62	60.98	66.36	66.36	66.36	66.36	66.36
	Caa-C	10	20.00	30.00	30.00	30.00	30.00	30.00	30.00	30.00	58.00	58.00	58.00	58.00	58.00	58.00	58.00	58.00	58.00	58.00	58.00	58.00
	IG	1,324	0.00	0.25	0.51	1.33	1.90	2.20	2.42	2.42	2.55	2.55	2.68	2.68	2.97	3.27	4.17	4.79	5.11	5.11	5.11	5.11
	SG	740	4.23	8.13	13.58	22.07	30.82	34.18	37.13	37.87	39.51	39.97	41.01	42.20	42.85	45.59	47.75	51.65	52.50	52.50	52.50	52.50
	All Rated	2,064	1.51	3.03	5.01	8.20	11.08	12.18	13.06	13.24	13.70	13.80	14.11	14.33	14.67	15.36	16.42	17.52	17.90	17.90	17.90	17.90

Exhibit 26 - Cumulative Issuer-Weighted Default Rates by Annual Cohort, 1970-2004

Cohort Year	Cohort Rating	Gross Denominator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1988	Aaa	128	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Aa	334	0.00	0.00	0.34	0.34	0.34	0.34	0.34	0.82	0.82	0.82	0.82	0.82	1.36	1.92	2.49	2.49	2.49
	A	553	0.00	0.19	0.79	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.83	2.16	2.86	2.86
	Baa	332	0.00	0.33	1.03	2.48	3.63	4.48	4.48	4.48	4.48	5.04	5.04	5.65	6.28	8.22	9.56	10.27	10.27
	Ba	508	1.24	6.51	12.87	20.66	23.71	26.93	27.85	28.86	29.98	31.26	31.98	32.78	34.49	39.11	43.17	43.17	43.17
	B	329	6.36	13.43	26.06	37.28	41.45	46.54	47.40	50.44	50.44	51.71	57.65	59.35	62.96	64.96	69.48	69.48	69.48
	Caa-C	11	28.57	28.57	28.57	28.57	28.57	28.57	28.57	64.29	64.29	64.29	64.29	64.29	64.29	64.29	64.29	64.29	64.29
	IG	1,347	0.00	0.16	0.66	1.17	1.45	1.65	1.65	1.76	1.76	1.89	1.89	2.02	2.29	3.11	3.68	4.12	4.12
	SG	848	3.59	9.50	18.17	27.13	30.53	34.35	35.23	37.21	37.95	39.19	41.55	42.60	44.85	48.51	52.54	52.54	52.54
	All Rated	2,195	1.37	3.63	6.98	10.22	11.42	12.62	12.85	13.43	13.60	13.97	14.45	14.75	15.36	16.63	17.74	18.08	18.08
1989	Aaa	143	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Aa	327	0.00	0.00	0.00	0.00	0.00	0.00	0.46	0.46	0.46	0.46	0.46	0.99	1.52	2.06	2.06	2.06	2.06
	A	587	0.00	0.18	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	1.41	1.71	2.02	2.02	2.02
	Baa	348	0.59	1.22	1.87	2.90	2.90	2.90	2.90	2.90	3.37	3.37	3.90	4.43	6.64	8.37	9.59	9.59	9.59
	Ba	489	2.98	9.99	18.13	20.83	24.02	24.43	25.35	26.89	28.63	30.60	31.33	33.70	37.95	43.55	43.55	43.55	43.55
	B	365	8.95	23.14	33.59	38.66	44.32	46.37	50.37	50.37	52.53	56.48	57.96	61.13	62.86	66.67	66.67	66.67	66.67
	Caa-C	12	25.00	41.67	53.33	53.33	53.33	53.33	53.33	53.33	53.33	53.33	53.33	53.33	53.33	53.33	53.33	53.33	53.33
	IG	1,405	0.15	0.37	0.69	0.94	0.94	0.94	1.04	1.04	1.16	1.16	1.28	1.52	2.52	3.17	3.57	3.57	3.57
	SG	866	5.79	15.88	24.97	28.54	32.60	33.60	35.60	36.56	38.38	40.90	41.84	44.36	47.62	52.38	52.38	52.38	52.38
	All Rated	2,271	2.27	6.05	9.32	10.62	11.83	12.11	12.70	12.94	13.43	13.95	14.22	14.87	16.21	17.51	17.82	17.82	17.82
1990	Aaa	160	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Aa	370	0.00	0.00	0.00	0.00	0.00	0.38	0.38	0.38	0.38	0.38	0.38	0.84	1.30	1.30	1.30	1.30	1.30
	A	614	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.26	0.53	0.81	1.10	1.10	1.10	1.10
	Baa	342	0.00	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	1.13	1.65	4.35	6.57	7.76	7.76	7.76	7.76
	Ba	473	3.35	11.85	14.40	17.39	18.16	19.43	20.36	22.39	24.09	24.71	27.38	32.39	36.30	36.30	36.30	36.30	36.30
	B	384	16.18	27.57	34.52	39.65	41.34	44.05	44.86	46.83	50.58	53.62	56.99	58.90	65.39	65.39	65.39	65.39	65.39
	Caa-C	18	58.82	67.06	67.06	67.06	67.06	100.00											
	IG	1,486	0.00	0.14	0.14	0.14	0.14	0.24	0.24	0.24	0.24	0.35	0.57	1.37	2.08	2.45	2.45	2.45	2.45
	SG	875	10.08	19.77	24.09	27.88	28.99	31.02	31.87	33.79	36.01	37.24	39.92	43.80	48.06	48.06	48.06	48.06	48.06
	All Rated	2,361	3.64	7.02	8.40	9.54	9.85	10.45	10.67	11.12	11.59	11.92	12.60	13.93	15.21	15.49	15.49	15.49	15.49

Exhibit 26 - Cumulative Issuer-Weighted Default Rates by Annual Cohort, 1970-2004

Cohort Year	Cohort Rating	Gross Denominator	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1991	Aaa	145	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Aa	415	0.00	0.00	0.00	0.00	0.32	0.32	0.32	0.32	0.32	0.32	0.72	1.13	1.13	1.13
	A	609	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.26	0.52	0.79	1.07	1.07
	Baa	374	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.69	1.12	3.31	5.14	6.11	6.11
	Ba	375	5.35	6.63	8.47	8.89	9.84	10.86	12.56	14.47	15.89	18.96	23.92	27.54	27.54	28.58
	B	329	14.56	23.68	30.52	32.71	36.61	37.39	40.17	43.58	46.27	49.29	51.04	59.04	59.04	59.04
	Caa-C	20	36.84	36.84	36.84	36.84	47.37	47.37	47.37	47.37	47.37	47.37	73.68	73.68	73.68	73.68
	IG	1,543	0.07	0.07	0.07	0.07	0.15	0.15	0.15	0.15	0.25	0.46	1.19	1.84	2.18	2.18
	SG	724	10.40	15.11	19.06	20.22	22.60	23.49	25.49	27.81	29.57	32.44	36.63	41.26	41.26	41.94
	All Rated	2,267	3.28	4.66	5.75	6.06	6.70	6.91	7.34	7.79	8.19	8.84	10.11	11.35	11.63	11.72
1992	Aaa	136	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Aa	424	0.00	0.00	0.00	0.29	0.29	0.29	0.29	0.29	0.29	0.65	0.65	0.65	0.65	0.65
	A	694	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.22	0.45	0.93	1.18	1.18	1.18
	Baa	393	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.38	0.77	2.79	4.09	5.01	5.01	5.01
	Ba	356	0.30	1.01	1.01	1.90	2.39	3.46	5.26	6.59	8.77	13.39	17.53	17.53	18.48	18.48
	B	260	9.03	17.54	20.90	25.63	27.99	30.76	34.14	36.75	42.63	44.32	52.00	52.00	52.00	52.00
	Caa-C	24	26.67	32.10	32.10	40.09	40.09	52.07	52.07	52.07	52.07	76.04	76.04	76.04	76.04	76.04
	IG	1,647	0.00	0.00	0.00	0.08	0.08	0.08	0.08	0.17	0.35	1.01	1.51	1.82	1.82	1.82
	SG	640	4.85	8.93	10.29	12.88	14.04	15.99	18.22	19.91	23.14	27.12	32.00	32.00	32.64	32.64
	All Rated	2,287	1.33	2.39	2.72	3.37	3.62	4.01	4.43	4.80	5.47	6.65	7.80	8.06	8.16	8.16
1993	Aaa	109	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Aa	439	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	A	740	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.19	0.59	1.00	1.21	1.21	1.21	1.21
	Baa	462	0.00	0.00	0.25	0.25	0.25	0.54	1.14	1.75	3.01	4.01	4.71	4.71	4.71	4.71
	Ba	384	0.56	0.56	2.64	3.40	4.67	6.11	7.17	8.92	12.61	15.92	15.92	15.92	15.92	15.92
	B	258	5.71	9.90	15.26	17.10	19.99	22.60	27.53	34.02	37.72	43.20	44.71	46.44	46.44	46.44
	Caa-C	23	28.57	28.57	42.18	50.44	50.44	50.44	50.44	50.44	75.22	75.22	75.22	75.22	75.22	75.22
	IG	1,750	0.00	0.00	0.07	0.07	0.07	0.14	0.30	0.54	1.03	1.46	1.73	1.73	1.73	1.73
	SG	665	3.52	5.15	8.83	10.18	12.00	13.79	16.13	19.44	23.40	27.26	27.73	28.24	28.24	28.24
	All Rated	2,415	0.95	1.38	2.33	2.65	3.04	3.46	4.02	4.80	5.88	6.87	7.17	7.25	7.25	7.25

Exhibit 26 - Cumulative Issuer-Weighted Default Rates by Annual Cohort, 1970-2004

Cohort Year	Cohort Rating	Gross Denominator	1	2	3	4	5	6	7	8	9	10	11
1994	Aaa	103	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Aa	397	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	A	880	0.00	0.00	0.00	0.00	0.00	0.00	0.15	0.46	0.78	0.95	0.95
	Baa	534	0.00	0.20	0.20	0.42	0.65	1.59	2.07	3.34	4.14	4.72	4.72
	Ba	435	0.24	1.81	2.09	3.04	4.84	7.26	9.93	12.28	15.32	15.89	15.89
	B	335	3.82	9.18	12.36	14.21	18.27	23.16	29.34	34.45	39.99	43.03	46.49
	Caa-C	41	5.13	13.75	24.31	24.31	24.31	30.90	30.90	56.02	56.02	56.02	56.02
	IG	1,914	0.00	0.06	0.06	0.12	0.18	0.44	0.64	1.12	1.48	1.71	1.71
	SG	811	1.95	5.38	7.28	8.54	11.08	14.49	18.28	22.07	25.86	27.28	28.45
	All Rated	2,725	0.57	1.57	2.08	2.44	3.07	3.99	4.89	6.00	6.99	7.42	7.62
1995	Aaa	104	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Aa	459	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	A	968	0.00	0.00	0.00	0.00	0.00	0.13	0.39	0.80	0.94	0.94	0.94
	Baa	553	0.00	0.00	0.00	0.42	1.29	1.74	3.15	3.64	3.90	3.90	3.90
	Ba	453	0.69	0.94	2.05	3.62	6.43	9.12	11.57	14.20	15.69	15.69	15.69
	B	452	4.81	7.33	10.21	13.93	18.35	24.19	32.91	39.67	43.73	45.71	45.71
	Caa-C	66	11.57	19.52	19.52	24.63	30.78	34.73	55.62	68.30	68.30	78.86	78.86
	IG	2,084	0.00	0.00	0.00	0.11	0.34	0.51	1.00	1.31	1.44	1.44	1.44
	SG	971	3.33	5.10	6.96	9.71	13.40	17.52	23.41	28.17	30.72	31.86	31.86
	All Rated	3,055	1.04	1.57	2.10	2.91	3.97	5.05	6.73	8.00	8.62	8.85	8.85
1996	Aaa	112	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Aa	481	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	A	1,071	0.00	0.00	0.00	0.00	0.11	0.34	0.58	0.71	0.71	0.71	0.71
	Baa	632	0.00	0.00	0.17	0.71	1.27	2.61	3.20	3.63	3.85	3.85	3.85
	Ba	475	0.00	0.72	2.35	5.36	7.63	11.06	13.30	15.00	15.00	15.00	15.00
	B	514	1.44	4.45	9.25	13.04	19.10	26.80	34.55	38.66	41.66	41.66	41.66
	Caa-C	76	13.99	20.24	27.41	40.61	46.55	67.11	72.17	72.17	86.08	86.08	86.08
	IG	2,296	0.00	0.00	0.05	0.20	0.40	0.88	1.15	1.33	1.39	1.39	1.39
	SG	1,065	1.67	3.86	7.37	11.39	15.53	21.54	26.27	28.92	30.41	30.41	30.41
	All Rated	3,361	0.52	1.17	2.17	3.31	4.47	6.24	7.51	8.19	8.53	8.53	8.53

Exhibit 26 - Cumulative Issuer-Weighted Default Rates by Annual Cohort, 1970-2004

Cohort Year	Cohort Rating	Gross Denominator	1	2	3	4	5	6	7	8
1997	Aaa	112	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Aa	536	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	A	1,133	0.00	0.00	0.00	0.10	0.31	0.53	0.88	0.88
	Baa	731	0.00	0.14	0.73	1.34	2.47	3.14	3.50	3.69
	Ba	559	0.19	1.48	5.05	8.15	11.16	13.85	15.55	15.55
	B	592	2.12	7.45	11.99	17.82	27.97	35.14	38.91	41.99
	Caa-C	78	14.67	27.31	39.59	50.10	72.63	76.84	76.84	76.84
	IG	2,512	0.00	0.04	0.21	0.44	0.86	1.15	1.40	1.46
	SG	1,229	2.06	6.01	10.50	15.23	22.40	27.18	29.73	31.09
	All Rated	3,741	0.66	1.88	3.28	4.71	6.90	8.30	9.08	9.41
1998	Aaa	99	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
	Aa	587	0.00	0.00	0.00	0.00	0.00	0.00	0.22	
	A	1,154	0.00	0.00	0.09	0.29	0.49	0.81	0.81	
	Baa	870	0.12	0.48	0.98	2.02	3.23	3.52	3.83	
	Ba	670	0.63	2.88	6.04	10.00	13.03	15.01	15.57	
	B	740	4.26	11.62	18.96	29.92	37.29	42.05	46.12	
	Caa-C	114	15.09	31.32	40.96	62.67	67.54	71.60	74.18	
	IG	2,710	0.04	0.15	0.35	0.77	1.25	1.47	1.62	
	SG	1,524	3.45	9.17	14.81	23.31	28.51	31.91	34.23	
	All Rated	4,234	1.24	3.24	5.21	8.20	10.11	11.24	11.94	
1999	Aaa	89	0.00	0.00	0.00	0.00	0.00	0.00		
	Aa	627	0.00	0.00	0.00	0.00	0.00	0.20		
	A	1,173	0.00	0.00	0.18	0.37	0.67	0.67		
	Baa	979	0.10	0.63	1.40	2.69	3.07	3.34		
	Ba	718	1.01	3.34	6.63	9.32	11.15	11.84		
	B	853	5.85	14.89	25.85	35.30	39.90	43.53		
	Caa-C	211	20.54	31.66	53.46	59.96	66.55	70.07		
	IG	2,868	0.04	0.22	0.56	1.07	1.32	1.45		
	SG	1,782	5.65	12.23	21.29	27.55	31.08	33.35		
	All Rated	4,650	2.16	4.69	8.17	10.66	11.96	12.72		

Exhibit 26 - Cumulative Issuer-Weighted Default Rates by Annual Cohort, 1970-2004

Cohort Year	Cohort Rating	Gross Denominator	1	2	3	4	5
2000	Aaa	102	0.00	0.00	0.00	0.00	0.00
	Aa	632	0.00	0.00	0.00	0.00	0.00
	A	1,237	0.00	0.17	0.35	0.63	0.73
	Baa	1,074	0.38	0.87	2.22	2.67	2.91
	Ba	582	0.89	2.79	4.65	6.02	6.02
	B	1,029	5.49	16.09	25.46	30.57	33.77
	Caa-C	267	20.04	43.80	54.63	61.86	66.68
	IG	3,045	0.13	0.38	0.92	1.18	1.31
	SG	1,878	6.14	15.87	23.02	27.11	29.36
	All Rated	4,923	2.41	6.17	9.08	10.65	11.43
2001	Aaa	99	0.00	0.00	0.00	0.00	
	Aa	697	0.00	0.00	0.00	0.00	
	A	1,287	0.16	0.41	0.67	0.67	
	Baa	1,082	0.19	1.68	1.99	2.22	
	Ba	524	1.57	3.48	4.88	4.88	
	B	980	9.36	19.75	25.58	28.91	
	Caa-C	256	34.37	46.85	57.97	62.51	
	IG	3,165	10.58	18.63	23.64	25.98	
	SG	1,760	10.58	10.58	10.58	10.58	
	All Rated	4,925	3.82	6.97	8.75	9.49	
2002	Aaa	108	0.00	0.00	0.00		
	Aa	726	0.00	0.00	0.00		
	A	1,301	0.16	0.33	0.33		
	Baa	1,187	1.21	1.49	1.59		
	Ba	543	1.54	3.21	3.90		
	B	760	4.97	10.66	13.29		
	Caa-C	296	30.30	43.41	50.88		
	IG	3,322	0.49	0.66	0.69		
	SG	1,599	8.45	13.95	16.53		
	All Rated	4,921	3.04	4.84	5.60		

Exhibit 26 - Cumulative Issuer-Weighted Default Rates by Annual Cohort, 1970-2004

Cohort Year	Cohort Rating	Gross Denominator	1	2
2003	Aaa	105	0.00	0.00
	Aa	683	0.00	0.00
	A	1,279	0.00	0.00
	Baa	1,160	0.00	0.00
	Ba	552	0.95	1.16
	B	752	2.66	4.07
	Caa-C	282	21.53	34.32
	IG	3,227	0.00	0.00
	SG	1,586	5.27	7.98
	All Rated	4,813	1.70	2.51
2004	Aaa	119	0.00	
	Aa	649	0.00	
	A	1,244	0.00	
	Baa	1,192	0.00	
	Ba	551	0.19	
	B	808	0.65	
	Caa-C	247	12.33	
	IG	3,204	0.00	
	SG	1,606	2.23	
	All Rated	4,810	0.72	

Recovery and Credit Loss Rates

Exhibit 27 - Annual Average Issuer-Weighted Defaulted Bond Recovery Rates by Seniority Class, 1982-2004

(Price per \$100 Par)

Default

Year	Sr. Secured	Sr. Unsecured	Sr. Subordinated	Subordinated	Jr. Subordinated	All bonds
1982	\$72.5	\$34.4	\$48.1	\$32.3	NA	\$35.0
1983	\$40.0	\$52.7	\$43.5	\$41.4	NA	\$50.1
1984	NA	\$49.4	\$67.9	\$44.3	NA	\$44.4
1985	\$83.6	\$60.2	\$30.9	\$42.7	\$48.5	\$39.9
1986	\$59.2	\$52.6	\$50.2	\$42.9	NA	\$44.3
1987	\$71.0	\$62.7	\$46.5	\$46.2	NA	\$61.7
1988	\$55.3	\$45.2	\$33.4	\$33.0	\$36.5	\$42.9
1989	\$46.5	\$43.8	\$33.1	\$26.8	\$16.9	\$32.8
1990	\$35.7	\$37.0	\$26.7	\$19.5	\$10.7	\$27.5
1991	\$50.1	\$38.9	\$43.8	\$24.1	\$7.8	\$39.1
1992	\$62.7	\$52.1	\$47.9	\$37.8	\$13.5	\$45.5
1993	NA	\$37.1	\$51.9	\$43.7	NA	\$48.0
1994	\$69.3	\$53.7	\$29.6	\$33.7	NA	\$44.5
1995	\$63.6	\$47.6	\$34.3	\$39.4	NA	\$45.8
1996	\$47.6	\$62.8	\$43.8	\$22.6	NA	\$43.6
1997	\$76.0	\$55.1	\$44.7	\$38.4	\$30.6	\$51.8
1998	\$51.8	\$39.5	\$44.2	\$14.1	\$62.0	\$40.4
1999	\$43.3	\$38.3	\$29.1	\$35.5	NA	\$37.6
2000	\$41.7	\$24.4	\$20.3	\$31.9	\$15.5	\$25.7
2001	\$41.7	\$23.1	\$20.9	\$15.9	\$47.0	\$34.3
2002	\$49.3	\$30.5	\$25.3	\$24.5	NA	\$34.6
2003	\$63.5	\$41.4	\$39.6	\$12.3	NA	\$43.1
2004	\$80.8	\$50.1	\$44.4	NA	NA	\$58.5
Mean	\$57.4	\$44.9	\$39.1	\$32.0	\$28.9	\$42.2
Median	\$55.3	\$45.2	\$43.5	\$33.4	\$23.7	\$43.1
Min	\$35.7	\$23.1	\$20.3	\$12.3	\$7.8	\$25.7
Max	\$83.6	\$62.8	\$67.9	\$46.2	\$62.0	\$61.7
StDev	\$14.3	\$11.2	\$11.4	\$10.5	\$18.9	\$8.7
N	251	1,207	478	485	22	2,443

Exhibit 28 - Average Issuer-Weighted Recovery Rates by Whole Letter Rating, 1982-2004

(Price per \$100 Par)

Sr. Unsecured

Rating	Years Prior to Rating				
	1	2	3	4	5
Aaa	NA	NA	NA	\$97.0	\$74.1
Aa	\$95.4	\$62.1	\$30.8	\$44.4	\$41.1
A	\$49.8	\$49.0	\$43.1	\$46.0	\$45.2
Baa	\$43.3	\$41.3	\$45.3	\$42.2	\$41.6
Ba	\$40.7	\$44.2	\$43.8	\$45.4	\$44.0
B	\$38.4	\$38.1	\$39.0	\$40.0	\$37.8
Caa-Ca	\$36.4	\$33.8	\$31.0	\$42.3	\$12.3
IG	\$46.0	\$44.4	\$44.1	\$44.3	\$43.2
SG	\$38.2	\$38.6	\$39.1	\$41.7	\$39.4
All	\$39.0	\$39.4	\$40.2	\$42.5	\$40.7

Credit Rating Transitions

Exhibit 29 - Annual Aggregate Rating* Migration Rates, 1983-2004

Cohort Year	Issuer Count	% Upgraded	% Downgraded	% Unchanged	% WR	D-U Ratio
1983	1,370	8.25	12.41	75.55	3.80	1.50
1984	1,422	12.87	11.74	72.93	2.46	0.91
1985	1,620	9.63	13.02	73.02	4.32	1.35
1986	1,834	8.51	17.23	68.16	6.11	2.03
1987	2,064	6.59	11.87	74.56	6.98	1.80
1988	2,195	6.61	13.26	73.90	6.24	2.01
1989	2,271	6.61	13.78	73.76	5.86	2.09
1990	2,361	4.96	19.06	68.95	7.03	3.85
1991	2,267	5.78	18.70	69.25	6.26	3.24
1992	2,287	6.69	17.10	67.77	8.44	2.56
1993	2,415	7.99	13.13	70.35	8.53	1.64
1994	2,725	7.60	7.30	79.38	5.72	0.96
1995	3,055	9.95	8.67	76.82	4.55	0.87
1996	3,361	11.34	7.53	75.75	5.39	0.66
1997	3,741	9.92	8.47	76.37	5.24	0.85
1998	4,234	8.57	17.67	68.21	5.55	2.06
1999	4,650	8.09	20.56	66.73	4.62	2.54
2000	4,923	11.01	12.09	72.27	4.63	1.10
2001	4,925	8.49	16.75	68.79	5.97	1.97
2002	4,921	5.10	21.19	66.35	7.36	4.16
2003	4,813	6.61	14.98	70.41	8.00	2.27
2004	4,810	13.58	8.77	70.69	6.96	0.65

Exhibit 30 - Annual Whole Letter Rating Migration Matrix, 2004

Cohort Rating	Issuer Count	Terminal Rating								WR
		Aaa	Aa	A	Baa	Ba	B	Caa-C	Default	
Aaa	119	97.47	1.52	0.00	0.00	0.00	0.00	0.00	0.00	1.01
Aa	649	0.42	91.75	2.24	0.14	0.00	0.00	0.00	0.00	5.45
A	1,244	0.08	3.02	89.02	3.09	0.00	0.00	0.00	0.00	4.80
Baa	1,192	0.00	0.08	6.57	84.72	3.04	0.16	0.00	0.00	5.42
Ba	551	0.00	0.00	0.17	12.98	74.74	3.98	0.17	0.17	7.79
B	808	0.00	0.12	0.12	0.24	9.65	73.58	6.03	0.60	9.65
Caa-C	247	0.00	0.00	0.00	0.00	0.40	9.49	62.45	11.07	16.60

Exhibit 31 - Average Annual Whole Letter Rating Migration Matrix, 1920-2004

Cohort Rating	Issuer Count	Terminal Rating								WR
		Aaa	Aa	A	Baa	Ba	B	Caa-C	Default	
Aaa	6,586	88.19	6.95	0.79	0.18	0.03	0.00	0.00	0.00	3.86
Aa	21,768	1.28	85.35	6.64	0.72	0.19	0.04	0.00	0.06	5.72
A	37,400	0.08	2.89	84.96	5.56	0.71	0.14	0.03	0.07	5.55
Baa	36,544	0.05	0.32	4.68	80.69	5.46	0.82	0.15	0.30	7.53
Ba	30,351	0.01	0.09	0.54	6.01	73.61	7.07	0.73	1.31	10.65
B	22,519	0.00	0.06	0.20	0.75	6.55	71.09	5.74	4.32	11.30
Caa-C	5,098	0.00	0.02	0.04	0.16	0.96	6.53	67.30	14.36	10.63

Exhibit 32 - Average Annual Whole Letter Rating Migration Matrix, 1970-2004

Cohort Rating	Issuer Count	Terminal Rating								WR
		Aaa	Aa	A	Baa	Ba	B	Caa-C	Default	
Aaa	3,179	89.48	7.05	0.75	0.00	0.03	0.00	0.00	0.00	2.69
Aa	11,310	1.07	88.41	7.35	0.25	0.07	0.01	0.00	0.00	2.83
A	22,981	0.05	2.32	88.97	4.85	0.46	0.12	0.01	0.02	3.19
Baa	18,368	0.05	0.23	5.03	84.50	4.60	0.74	0.15	0.16	4.54
Ba	12,702	0.01	0.04	0.46	5.28	78.88	6.48	0.50	1.16	7.19
B	10,794	0.01	0.03	0.12	0.40	6.18	77.45	2.93	6.03	6.85
Caa-C	2,091	0.00	0.00	0.00	0.52	1.57	4.00	62.68	23.12	8.11

Exhibit 33 - Average 5-Year Whole Letter Rating Migration Matrix, 1970-2004

Cohort Rating	Issuer Count	Terminal Rating								WR
		Aaa	Aa	A	Baa	Ba	B	Caa-C	Default	
Aaa	2,792	56.88	23.78	5.58	0.46	0.4	0.04	0.08	0.11	12.67
Aa	8,751	4.16	53.86	23.13	3.58	0.9	0.29	0.02	0.21	13.84
A	18,268	0.25	8.15	57.83	14.2	2.95	0.82	0.16	0.43	15.22
Baa	14,116	0.24	1.51	15.64	47.05	9.58	2.65	0.47	1.72	21.14
Ba	10,923	0.08	0.25	2.98	12.53	32.12	11.1	1.07	8.12	31.76
B	7,720	0.05	0.08	0.51	2.82	12.55	29.56	2.31	20.58	31.54
Caa-C	1,073	0	0	0	3.03	5.62	7.06	15.12	42.85	26.31

Exhibit 34 - Average 10-Year Whole Letter Rating Migration Matrix, 1970-2004

Cohort Rating	Issuer Count	Terminal Rating								WR
		Aaa	Aa	A	Baa	Ba	B	Caa-C	Default	
Aaa	2,278	32.38	30.47	10.41	2.97	0.76	0.1	0.05	0.6	22.26
Aa	5,888	4.83	30.2	28.25	7.94	2.32	0.58	0.09	0.78	25.01
A	12,500	0.36	10.4	38.29	15.66	4.32	1.53	0.24	1.24	27.96
Baa	9,830	0.21	2.38	17.33	26.84	7.81	3.08	0.38	3.63	38.35
Ba	7,919	0.2	0.81	5.26	11.37	11.3	6.82	0.7	13.67	49.87
B	3,992	0.06	0.03	1.62	3.98	8.58	9.41	0.75	27.39	48.18
Caa-C	327	0	0	0	4.49	1.92	1.85	2.14	50.42	39.17

Exhibit 35 - Annual Alphanumeric Rating Migration Matrix, 2004

Cohort Rating	Issuer Count	Terminal Rating																		
		Aaa	Aa1	Aa2	Aa3	A1	A2	A3	Baa1	Baa2	Baa3	Ba1	Ba2	Ba3	B1	B2	B3	Caa-C	Default	WR
Aaa	119	97.47	1.01	0.51	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.01
Aa1	152	1.78	74.56	13.61	4.14	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.92
Aa2	159	0.00	2.20	89.01	2.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6.59
Aa3	338	0.00	0.82	4.40	85.44	1.10	3.30	0.00	0.27	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.67
A1	340	0.28	0.00	0.85	6.76	77.18	9.01	0.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.35
A2	417	0.00	0.00	0.68	1.14	8.45	79.45	4.79	0.68	0.68	0.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.42
A3	486	0.00	0.20	0.40	0.20	9.20	2.60	75.60	3.80	1.80	0.60	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.60
Baa1	422	0.00	0.00	0.00	0.00	0.23	3.51	9.37	75.64	6.56	0.47	0.00	0.00	0.00	0.00	0.23	0.00	0.00	0.00	3.98
Baa2	407	0.00	0.00	0.00	0.00	0.48	0.00	4.33	6.01	77.40	4.33	1.44	0.00	0.24	0.24	0.00	0.00	0.00	0.00	5.53
Baa3	363	0.00	0.00	0.00	0.27	0.27	0.27	0.53	8.29	8.56	66.84	5.88	1.60	0.53	0.00	0.00	0.00	0.00	0.00	6.95
Ba1	179	0.00	0.00	0.00	0.00	0.53	0.00	0.00	1.60	8.51	17.55	52.13	7.98	4.79	0.53	0.53	0.00	0.00	0.00	5.85
Ba2	150	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.63	1.25	10.00	9.38	61.88	5.63	1.88	0.00	0.63	0.00	0.00	8.75
Ba3	222	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.43	0.00	1.30	5.65	16.09	59.57	3.48	3.04	0.87	0.43	0.43	8.70
B1	283	0.00	0.00	0.00	0.35	0.00	0.00	0.00	0.00	0.00	0.35	0.35	9.69	14.88	52.60	5.54	3.81	1.73	0.00	10.73
B2	318	0.00	0.00	0.00	0.00	0.00	0.30	0.00	0.00	0.00	0.30	0.30	0.00	2.12	11.82	60.30	8.79	6.36	0.61	9.09
B3	207	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.76	11.43	61.90	11.43	1.43	9.05
Caa-C	246	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.40	0.00	0.00	1.58	2.77	5.14	62.45	11.07	16.60

Exhibit 36 - Average Annual Alphanumeric Rating Migration Matrix, 1983-2004

Cohort Rating	Issuer Count	Terminal Rating																		
		Aaa	Aa1	Aa2	Aa3	A1	A2	A3	Baa1	Baa2	Baa3	Ba1	Ba2	Ba3	B1	B2	B3	Caa-C	Default	WR
Aaa	2,433	86.31	5.92	2.79	0.41	0.58	0.21	0.08	0.00	0.00	0.00	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.90
Aa1	1,907	1.99	78.03	8.23	6.40	1.47	0.26	0.05	0.10	0.00	0.00	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.72
Aa2	3,134	0.64	3.19	78.43	8.81	3.13	1.24	0.48	0.10	0.06	0.00	0.00	0.00	0.03	0.03	0.00	0.00	0.00	0.00	3.99
Aa3	5,092	0.10	0.59	3.69	78.48	8.92	2.99	0.67	0.22	0.10	0.08	0.02	0.02	0.04	0.00	0.00	0.00	0.00	0.00	4.28
A1	6,026	0.05	0.10	0.46	5.58	77.53	7.88	2.89	0.70	0.20	0.07	0.25	0.15	0.03	0.07	0.05	0.00	0.00	0.00	4.10
A2	7,077	0.04	0.04	0.34	0.76	5.26	77.26	7.60	2.95	0.88	0.47	0.28	0.07	0.14	0.08	0.07	0.00	0.06	0.03	3.87
A3	5,966	0.05	0.12	0.08	0.18	1.83	6.82	74.15	6.89	3.37	1.26	0.45	0.13	0.15	0.17	0.03	0.05	0.07	0.03	4.37
Baa1	4,736	0.04	0.04	0.13	0.17	0.23	2.11	6.63	72.78	7.79	3.13	0.95	0.44	0.21	0.65	0.06	0.08	0.25	0.17	4.48
Baa2	5,000	0.06	0.10	0.04	0.16	0.22	0.62	3.46	5.74	73.52	7.08	1.72	0.54	0.68	0.52	0.32	0.12	0.36	0.12	4.94
Baa3	3,991	0.05	0.00	0.03	0.08	0.18	0.40	0.68	3.08	8.22	69.58	6.16	2.68	1.73	0.65	0.45	0.23	0.70	0.40	5.21
Ba1	3,330	0.03	0.00	0.00	0.03	0.24	0.21	0.54	0.96	3.12	8.62	64.32	5.14	4.62	1.59	1.14	0.81	0.90	0.63	7.99
Ba2	2,701	0.00	0.00	0.04	0.04	0.04	0.19	0.11	0.37	0.81	2.48	8.96	62.05	7.74	2.63	3.11	1.07	1.33	0.59	9.74
Ba3	4,208	0.00	0.02	0.02	0.00	0.05	0.14	0.14	0.21	0.21	0.67	2.59	5.77	64.97	6.70	5.13	2.83	2.26	2.14	8.58
B1	4,747	0.02	0.00	0.02	0.02	0.06	0.11	0.08	0.06	0.23	0.21	0.48	2.99	6.07	63.37	8.59	4.93	4.59	2.91	8.09
B2	3,224	0.00	0.00	0.03	0.03	0.03	0.03	0.12	0.22	0.09	0.25	0.34	0.74	1.83	7.20	60.67	8.59	10.33	5.68	8.65
B3	2,601	0.00	0.00	0.08	0.00	0.04	0.08	0.08	0.12	0.12	0.15	0.12	0.38	0.65	3.46	5.77	60.21	17.88	10.27	9.30
Caa-C	2,079	0.00	0.00	0.00	0.00	0.00	0.00	0.05	0.00	0.05	0.05	0.24	0.00	0.43	0.77	1.06	4.71	72.87	20.97	11.74

Exhibit 37 - Average 5-Year Alphanumeric Rating Migration Matrix, 1983-2004

Cohort Rating	Issuer Count	Terminal Rating																		
		Aaa	Aa1	Aa2	Aa3	A1	A2	A3	Baa1	Baa2	Baa3	Ba1	Ba2	Ba3	B1	B2	B3	Caa-C	Default	WR
Aaa	2,002	48.58	10.60	9.99	5.13	3.55	1.32	1.06	0.41	0.05	0.00	0.00	0.00	0.06	0.07	0.00	0.00	0.14	0.20	18.84
Aa1	1,366	5.27	28.75	14.61	15.36	7.84	3.23	1.95	2.73	1.14	0.30	0.15	0.38	0.15	0.00	0.00	0.00	0.00	0.26	17.89
Aa2	2,389	2.29	4.98	29.45	17.58	10.44	8.09	3.29	2.03	0.87	0.52	0.32	0.33	0.23	0.35	0.00	0.00	0.00	0.31	18.92
Aa3	3,625	0.68	1.94	7.09	32.75	15.46	10.68	4.85	2.89	0.91	0.58	0.33	0.19	0.28	0.34	0.00	0.00	0.08	0.29	20.67
A1	4,642	0.39	1.01	2.24	10.37	34.36	13.56	8.63	4.04	2.20	1.27	0.81	0.15	0.37	0.55	0.23	0.16	0.14	0.61	18.91
A2	5,223	0.10	0.42	0.85	2.97	10.07	33.14	12.84	7.02	4.43	2.48	1.26	0.92	0.81	0.53	0.26	0.06	0.19	0.61	21.05
A3	4,100	0.03	0.24	0.35	2.33	4.73	15.50	25.19	10.03	7.66	4.46	2.13	1.36	1.80	0.88	0.36	0.10	0.18	0.51	22.17
Baa1	3,145	0.19	0.23	0.45	0.63	2.48	6.86	13.47	25.13	11.34	7.05	2.07	1.73	1.48	1.13	0.67	0.37	0.44	1.25	23.04
Baa2	3,360	0.25	0.51	0.24	0.92	1.59	3.75	7.09	11.01	28.69	9.63	2.80	1.48	1.45	1.63	1.08	0.58	0.98	1.56	24.75
Baa3	2,601	0.40	0.05	0.29	0.15	0.63	2.08	3.97	7.60	14.34	22.23	6.00	4.52	3.69	2.28	1.02	0.70	0.57	3.03	26.44
Ba1	2,556	0.09	0.10	0.08	0.09	0.64	1.38	2.51	2.91	7.12	8.91	16.10	4.26	4.99	3.14	2.95	1.61	1.26	5.69	36.16
Ba2	2,111	0.00	0.05	0.15	0.18	0.20	0.46	1.46	1.29	3.07	5.36	7.65	12.72	6.99	4.04	4.51	2.57	1.30	7.72	40.29
Ba3	3,404	0.00	0.05	0.02	0.04	0.00	0.23	0.51	0.82	0.92	1.99	3.53	4.27	14.50	5.82	5.70	3.46	2.07	14.97	41.11
B1	3,566	0.08	0.08	0.13	0.00	0.12	0.22	0.60	0.31	0.47	0.84	1.81	2.89	4.64	15.26	5.74	5.12	2.48	20.74	38.48
B2	1,931	0.09	0.00	0.00	0.15	0.00	0.09	0.25	0.34	0.66	0.63	1.36	2.36	2.61	7.58	11.40	5.81	4.22	26.00	36.47
B3	1,775	0.08	0.00	0.00	0.00	0.00	0.07	0.42	0.26	0.43	0.81	0.53	0.80	2.07	4.78	3.06	9.24	5.36	32.81	39.28
Caa-C	999	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.16	0.14	0.22	0.29	0.75	2.15	4.01	1.34	2.19	7.67	43.20	36.88

Appendix B - Methodology & Data

MOODY'S DEFINITION OF DEFAULT

Moody's definition of default includes three types of credit events:

1. A missed or delayed disbursement of interest and/or principal, including delayed payments made within a grace period;
2. Bankruptcy, administration, legal receivership, or other legal blocks (perhaps by regulators) to the timely payment of interest and/or principal; or
3. A distressed exchange occurs where: (i) the issuer offers bondholders a new security or package of securities that amount to a diminished financial obligation (such as preferred or common stock, or debt with a lower coupon or par amount, lower seniority, or longer maturity); or (ii) the exchange had the apparent purpose of helping the borrower avoid default.

The definition of a default is intended to capture events that change the relationship between the bondholder and bond issuer from the relationship which was originally contracted, and which subjects the bondholder to an economic loss. Technical defaults (covenant violations, etc.) are not included in Moody's definition of default.

ESTIMATED SENIOR RATINGS

The credit ratings used in Moody's annual default studies do not refer to the ratings of any specific debt obligations. Rather, they are notional, issuer-level ratings derived from each issuer's outstanding rated bonds. The resulting ratings are called estimated senior unsecured ratings or, more concisely, estimated senior ratings. Estimated senior ratings are derived by a statistical algorithm: first, the rating on each company's senior unsecured bonds is used; if an issuer does not have any rated senior unsecured bonds, the estimated senior rating is derived by statistically implying such a rating on the basis of rated subordinated or secured bonds, or senior implied ratings.

Deriving estimated senior ratings is necessary because Moody's credit ratings for corporate bonds are based on expected credit loss, which is comprised of (1) the expected probability of default, as well as (2) the expected severity (loss of principal and interest) in the event of default. Therefore, in order to calculate historical default rates by rating category, which are estimates of the expected default probability component, one must hold severity considerations constant. In most cases, estimated senior ratings will yield an assessment of default risk that is relatively unaffected by special considerations of collateral or of position within the capital structure.

DEFAULT RATE CALCULATION

The cumulative default rate calculation used in Moody's corporate bond default research since 1996 makes an adjustment to the denominator to reflect issuers whose ratings were withdrawn, and therefore not at risk of default over the measurement period. There are therefore three possible states that need to be modeled in the default rate: survival to the next time period, rating withdrawal, and default.

The cumulative default rate for a time horizon T is calculated as:

$$D(T) = 1 - \prod_{t=1}^T (1 - d_t)$$

Where d_t is the marginal default rate:

$$d_t = \frac{x_t}{n_t - w_t / 2}$$

Where x_t is the number of defaulters in year t , w_t is the number of rating withdrawals in year t , and where $n_t = n_{t-1} - x_{t-1} - w_{t-1}$, the number of issuers in the cohort at time t . When the time horizon T is equal to 1, the cumulative default rate and the marginal default rate are equal, and resulting rate represents the annual default rate.

RECOVERY RATES

Moody's measures recovery rates in this report using bid prices on defaulted debt obligations observed roughly 30 days after the date of default, specifically measured as a ratio of prices relative to par. This approach parallels common practice in the credit default swap market. Moreover, recovery rates measured in this way are most relevant for the many cash bond investors who liquidate their holdings shortly after default as required by their portfolio governance rules or their own investment objectives. For investors holding defaulted securities until ultimate resolution, prices observed shortly after default are generally accepted as the market's discounted expected ultimate recovery rates.⁶

6. The alternative approach - measuring realized recoveries - presents a number of estimation challenges, including estimation of the discount rate to apply cash-flows, assumptions about the value of certain payments made during the process, and estimates of the values of securities used to pay creditor claims, e.g., equity and derivative instruments. Absence of a market for these assets and securities makes such an assessment highly subjective.

DATA SOURCES

Moody's bases the results of this study on its proprietary database of ratings and defaults for corporate bond issuers. Municipal debt issuers, structured finance transactions, private placements, and issuers with only short-term debt ratings or only bank loan ratings are excluded unless otherwise noted. In total, Moody's data covers the credit experiences of over 16,000 corporate issuers that sold long-term public debt at some time between 1919 and 2003. As of January 1, 2005 over 5,000 corporate issuers held a Moody's long-term bond rating.

Moody's database of corporate defaults covers over 3,600 long-term bond defaults by issuers both rated and non-rated by Moody's. Additional data sources, such as Lehman Brothers index data, supplemented Moody's proprietary data in the construction of the aggregate dollar volume-weighted default rates. Defaulted bond pricing data was derived from Interactive Data Corp. (IDC), Bloomberg, and Reuters. The majority of these market quotes represent an actual bid on the debt instrument, although no trade may have occurred at that price. Over the 1982-2004 period, the dataset includes post-default prices for nearly 2,400 defaulted instruments issued by over 1,200 defaulting corporations.

Moody's makes the 1970-2004 credit rating, default, and recovery rate data used in this study available through its Default Risk Service (DRS) database.

Related Research

Special Comments:

[Measuring The Performance Of Corporate Bond Ratings, April 2003 \(77916\)](#)

[Predicting Default Rates: A Forecasting Model For Moody's Issuer-Based Default Rates, August 2003 \(47729\)](#)

[The Performance of Moody's Corporate Bond Ratings: December 2004 Quarterly Update, January 2005 \(91196\)](#)

[Determinants of Recovery Rates on Defaulted Bonds and Loans for North American Corporate Issuers: 1983-2003, December 2004 \(90593\)](#)

To access any of these reports, click on the entry above. Note that these references are current as of the date of publication of this report and that more recent reports may be available. All research may not be available to all clients.

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